
ELEC6095 Smart Grid

Course Overview and Renewable Energy Topologies

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Course overview



Course outline

1. understand the differences and challenges in the **future electric power grid**;
2. explain the basic **operating principle and control strategy of power systems and various type of renewable energy sources**;
3. explain **different renewable energy topologies and identify their benefits and applications**;
4. understand the functions and operation of **energy storage systems** and their applications in smart grid;
5. understand the concept of **integrated communication to optimize power system reliability**, asset utilization and security;
6. explain the operating principle and control strategy of various **smart devices/appliances** and energy saving control.

Class schedule

Course Content	Date
Part 1: Smart Grid Principles	
Introduction to Renewable Energy	20 th Jan., 2026
History and development of Modern Grid	27 th Jan., 2026
Challenges in Modern Grid Operation	3 rd Feb., 2026
Demand Response	10 th Feb., 2026
Lunar New Year holidays	17th Feb., 2026
Communication Perspectives for Smart Grid	24 th Feb., 2026
Part 2: Smart Grid Applications	
Grid Connection of Distributed Generation	3 rd March, 2026
Reading week	10th March, 2026
Integrating PV Systems & Load Management	17 th March, 2026
Intermittency of Renewable Energy	24 th April, 2026
Distributed Generation and Smart Grid	31 st April, 2026
Public holiday	7th April, 2026
Energy Storage in Smart Grid	13 th April, 2026

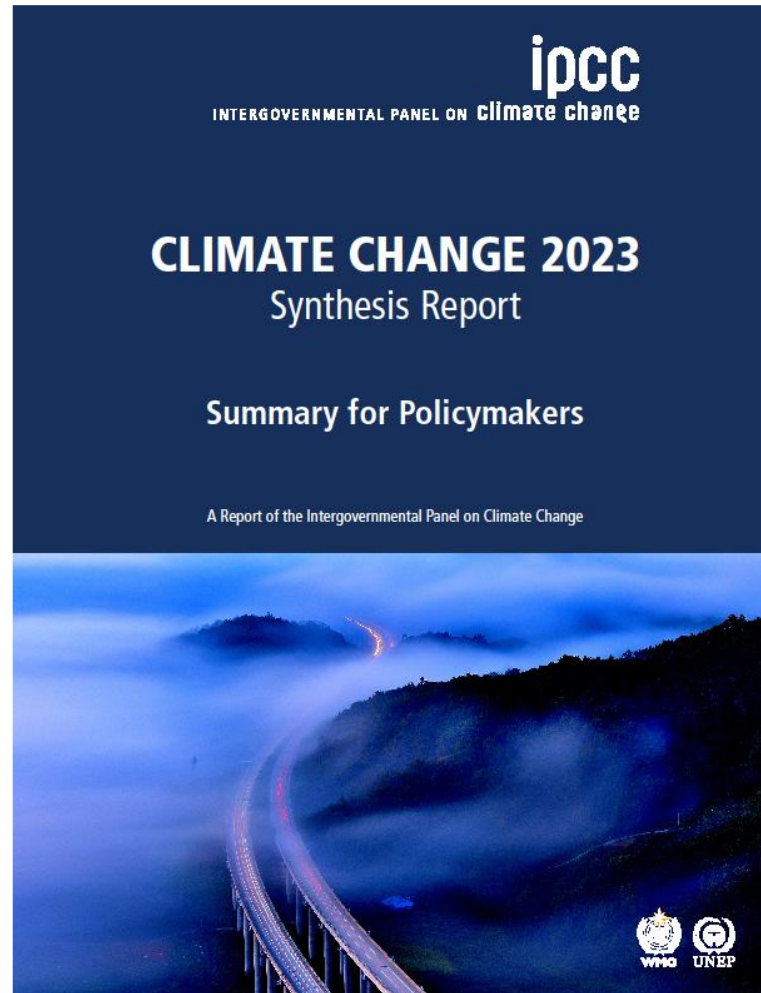
Allocation of marks

- ▶ 50% - 50% split between the course materials on *Principles* and *Applications*
- ▶ In the final exam, it will be separated into Section A and B, both section carries **EQUAL** weight
- ▶ The Exam will account for 60% of the marks
- ▶ **40% will be by coursework – 20% for each part from *Principles* and *Applications*:**
 1. 20% on a technical paper on the principles of Smart Grid to be submitted by 1st March.
 2. 20% on a technical report on the principles of Smart Grid to be submitted by 20th April.

Why we need Renewable Energy?

Environmental and pollution problems

Climate Change



IPCC AR6 Synthesis Report: Climate Change 2023:
<https://www.ipcc.ch/report/ar6/syr/>

The basis of Climate Change



https://youtu.be/G4H1N_yXBiA?si=DFmQduxPseZy_tbo

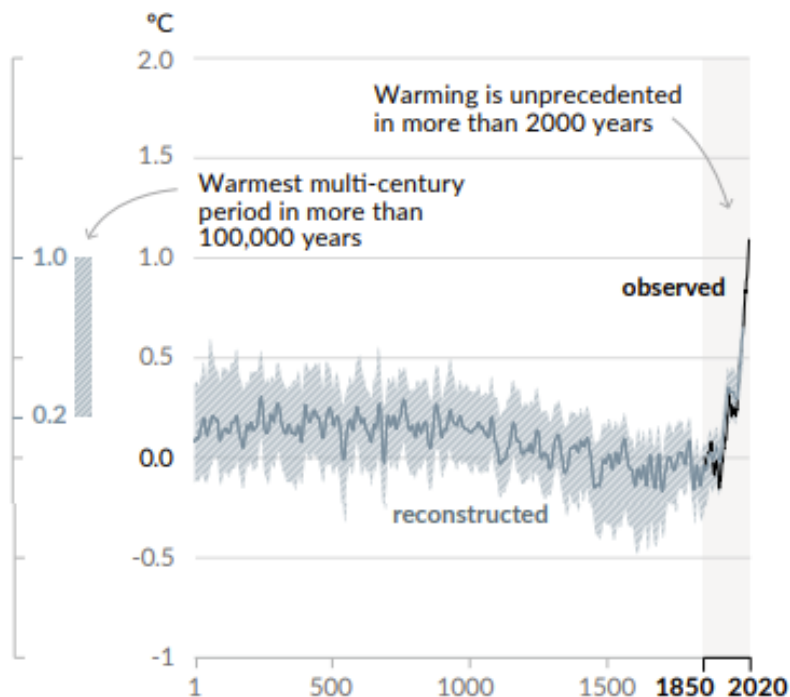
Climate Change

- ▶ Greenhouse gas (GHG) causes global rise of ambient temperature, changing weather pattern, rise of sea level and eventually affects the eco-system. According to [Intergovernmental Panel on Climate Change](#) (IPCC):
 - ▶ Global surface temperature has increased faster since 1970 than in any other 50-year period over at least the last 2000 years. Temperatures during the most recent decade (2011–2020) exceed those of the most recent multi-century warm period, around 6500 years ago;
 - ▶ Global mean sea level has risen faster since 1900 than over any preceding century in at least the last 3000 years;
 - ▶ In 2011–2020, annual average Arctic sea ice area reached its lowest level since at least 1850. Late summer Arctic sea ice area was smaller than at any time in at least the past 1000 years;
 - ▶ This change in earth surface temperature will lead to “dangerous climate change” (United Nations Framework Convention on Climate Change)
 - ▶ The following graphs are extracted from IPCC reports.

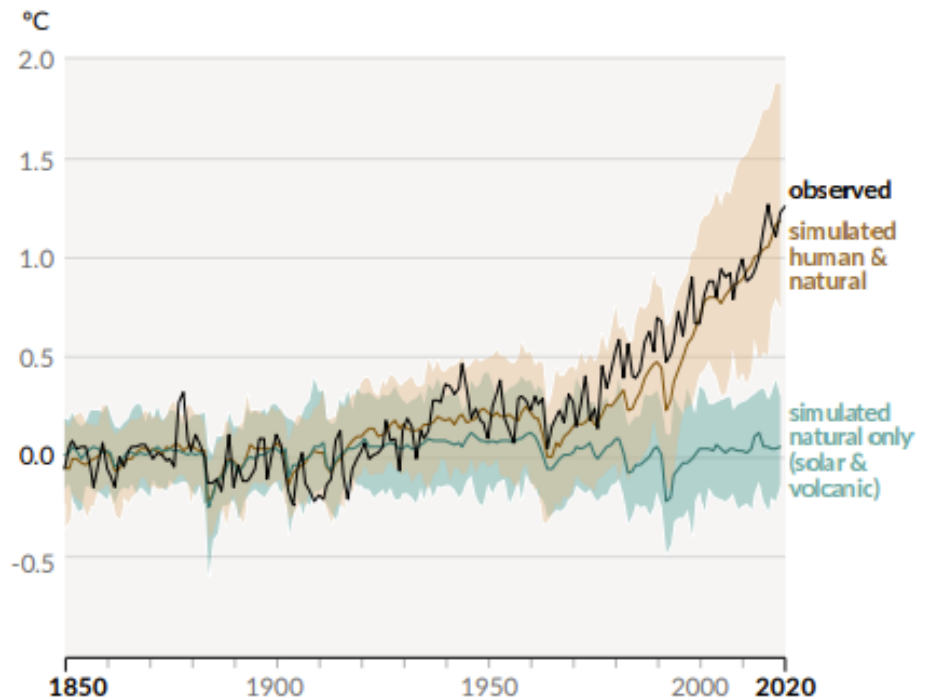
Global surface temperature

Changes in global surface temperature relative to 1850-1900

a) Change in global surface temperature (decadal average) as **reconstructed** (1-2000) and **observed** (1850-2020)



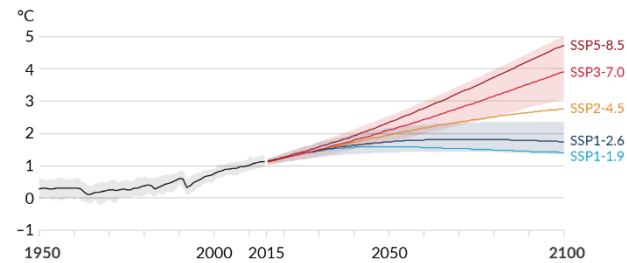
b) Change in global surface temperature (annual average) as **observed** and simulated using **human & natural** and **only natural** factors (both 1850-2020)



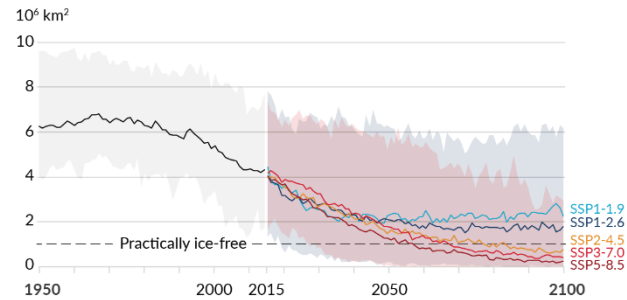
Source: <https://www.ipcc.ch/report/ar6/wg1/>

Human activities affect all the major climate system components, with some responding over decades and others over centuries

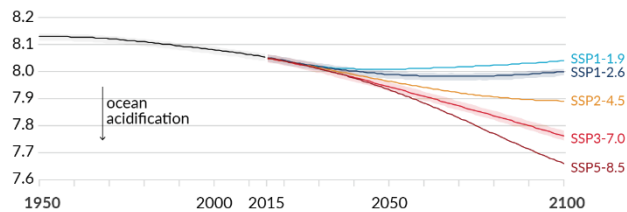
(a) Global surface temperature change relative to 1850–1900



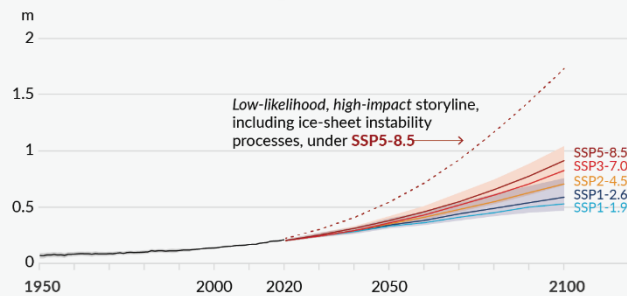
(b) September Arctic sea ice area



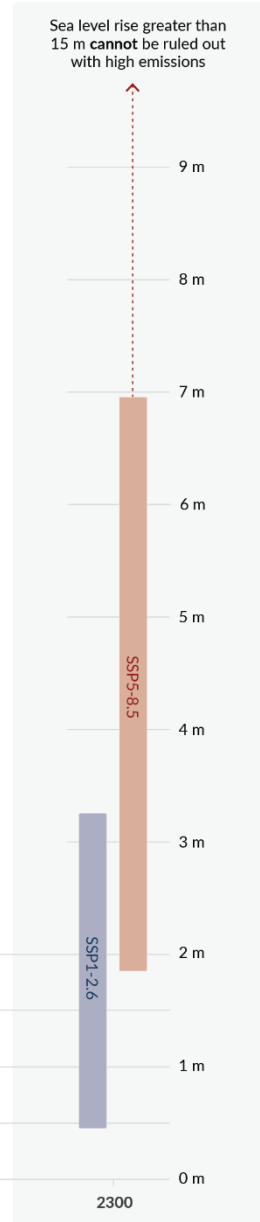
(c) Global ocean surface pH (a measure of acidity)



(d) Global mean sea level change relative to 1900



(e) Global mean sea level change in 2300 relative to 1900



Source:

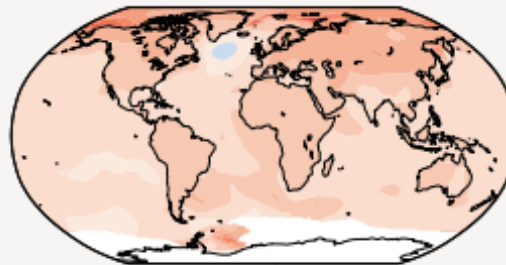
<https://www.ipcc.ch/report/ar6/wg1/>

Regional surface temperature changes

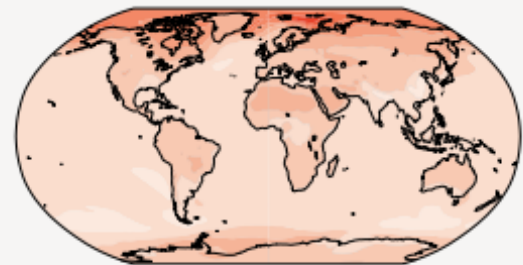
a) Annual mean temperature change (°C) at 1 °C global warming

Warming at 1 °C affects all continents and is generally larger over land than over the oceans in both observations and models. Across most regions, observed and simulated patterns are consistent.

Observed change per 1 °C global warming



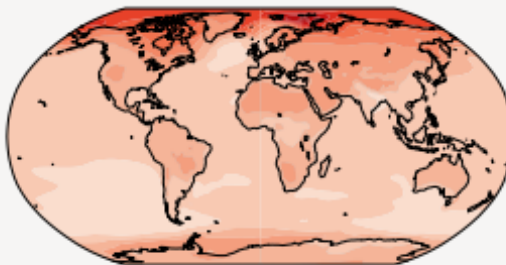
Simulated change at 1 °C global warming



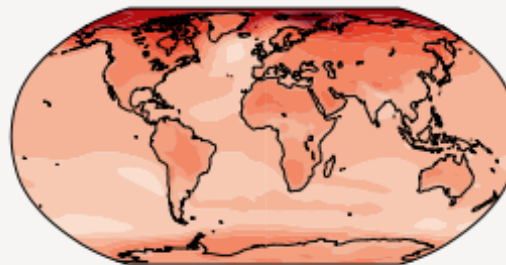
b) Annual mean temperature change (°C) relative to 1850-1900

Across warming levels, land areas warm more than oceans, and the Arctic and Antarctica warm more than the tropics.

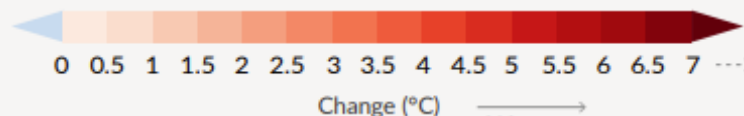
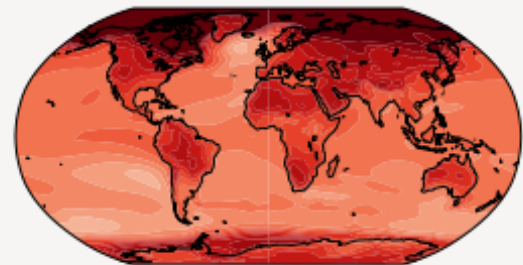
Simulated change at 1.5 °C global warming



Simulated change at 2 °C global warming

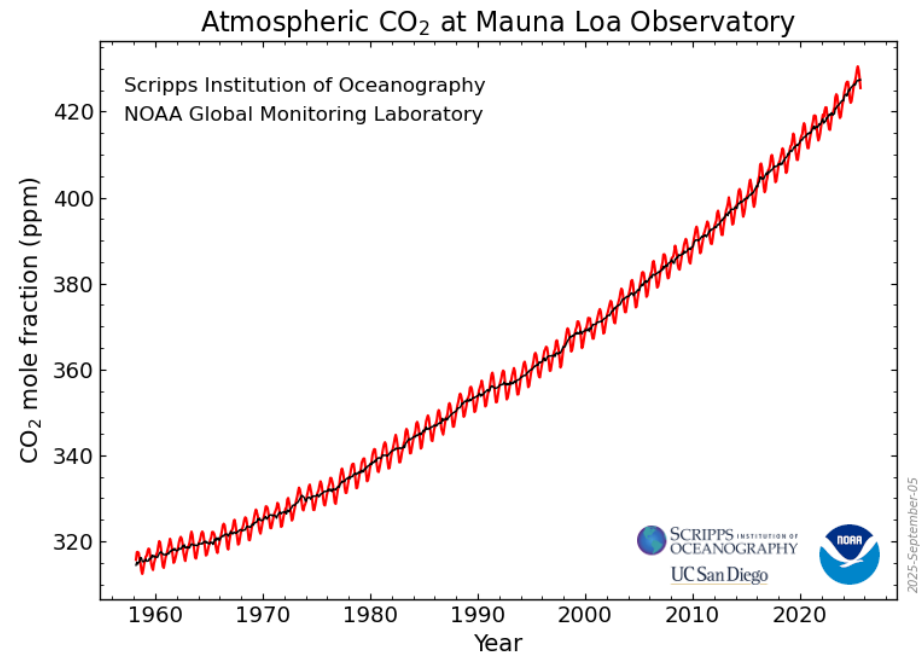


Simulated change at 4 °C global warming



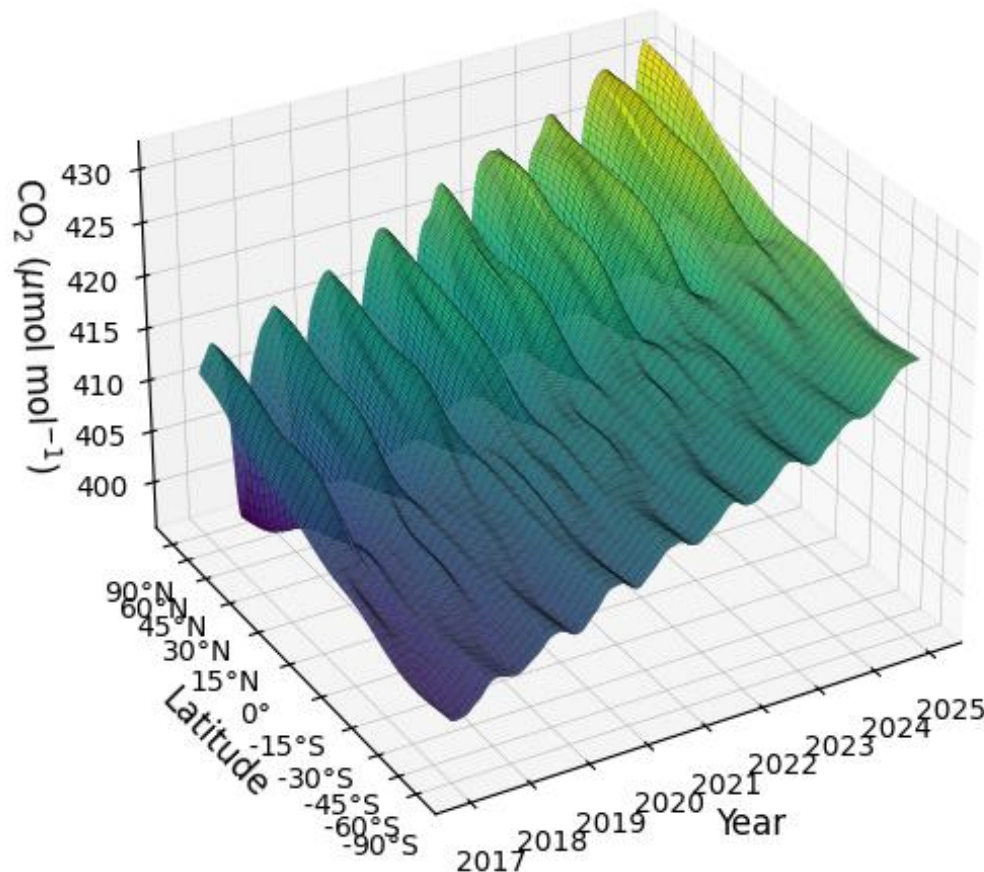
Rise of atmospheric CO₂

- ▶ On 10th May, 2025 it hits a record high of daily average 431.25 parts per million (ppm) CO₂ measured at the Mauna Loa Observatory.
- ▶ Prior to 1800, atmospheric CO₂ averaged about 280 ppm.
- ▶ The last time that the Earth experienced such a sustained CO₂ increase was between 17,000 and 11,000 years ago, when CO₂ levels increased by 80 ppm.



Source: <https://esrl.noaa.gov/gmd/ccgg/trends/>

Marine Boundary Layer (MBL*) Reference

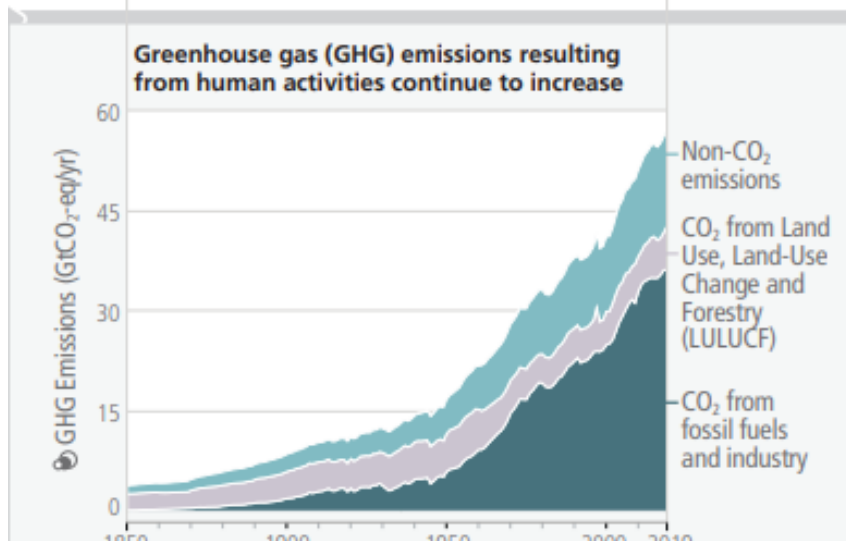
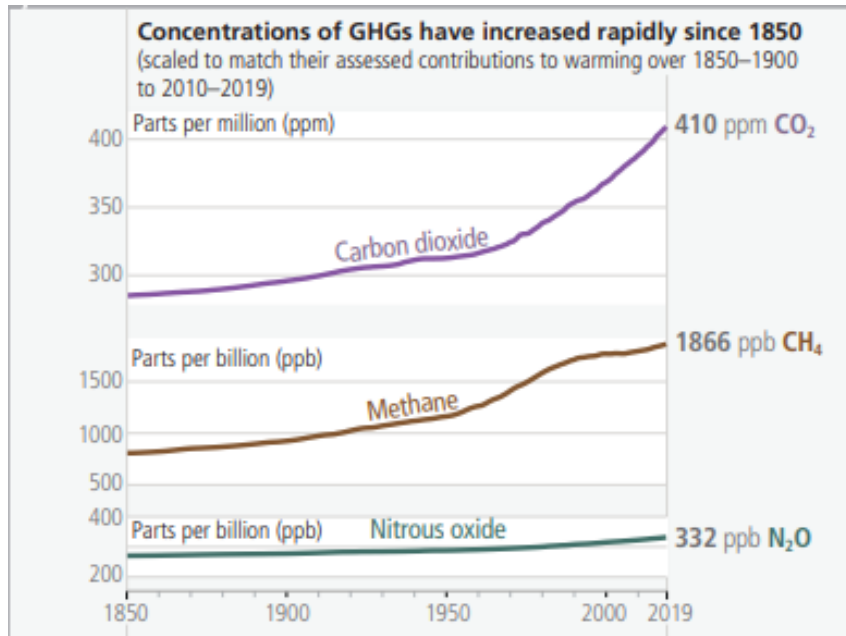


Source: <https://esrl.noaa.gov/gmd/ccgg/mbi/index.html>

* typically measured at remote marine sea level locations with prevailing onshore winds



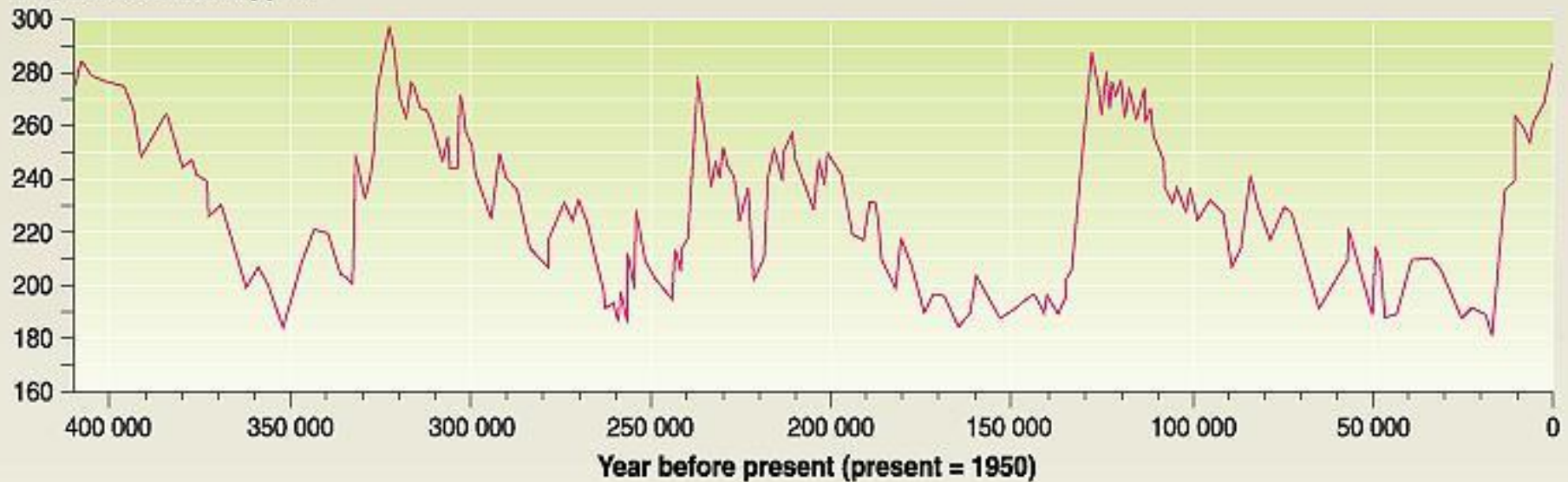
Global anthropogenic CO₂ emission



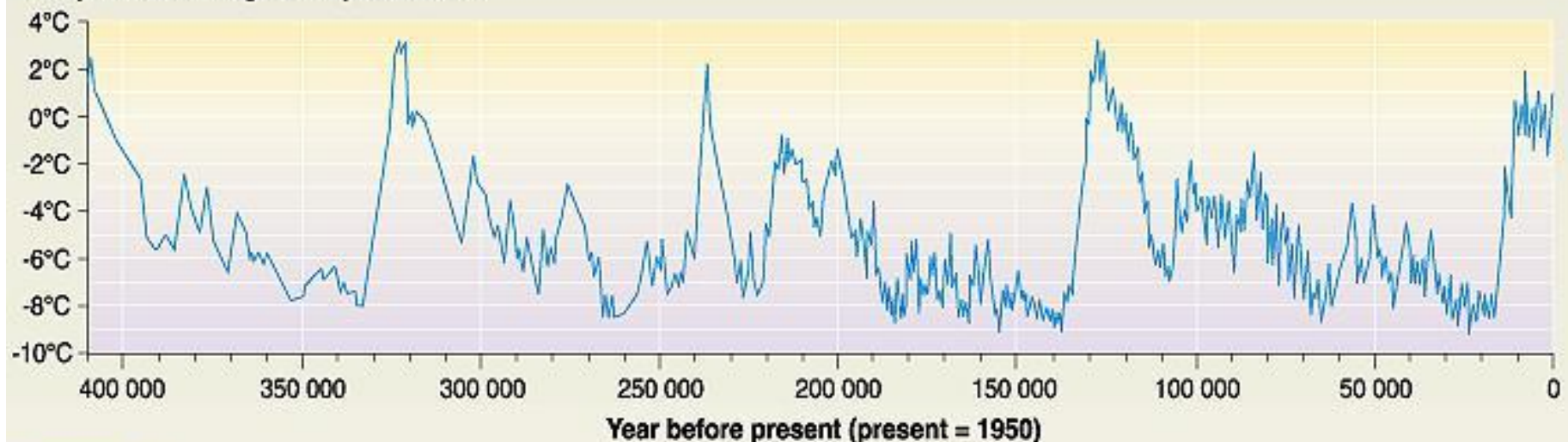
Source: *Climate Change 2023 Synthesis Report*
https://www.ipcc.ch/report/ar6/syr/downloads/report/IPCC_AR6_SYR_LongerReport.pdf

Temperature and CO₂ concentration in the atmosphere over the past 400 000 years (from the Vostok ice core)

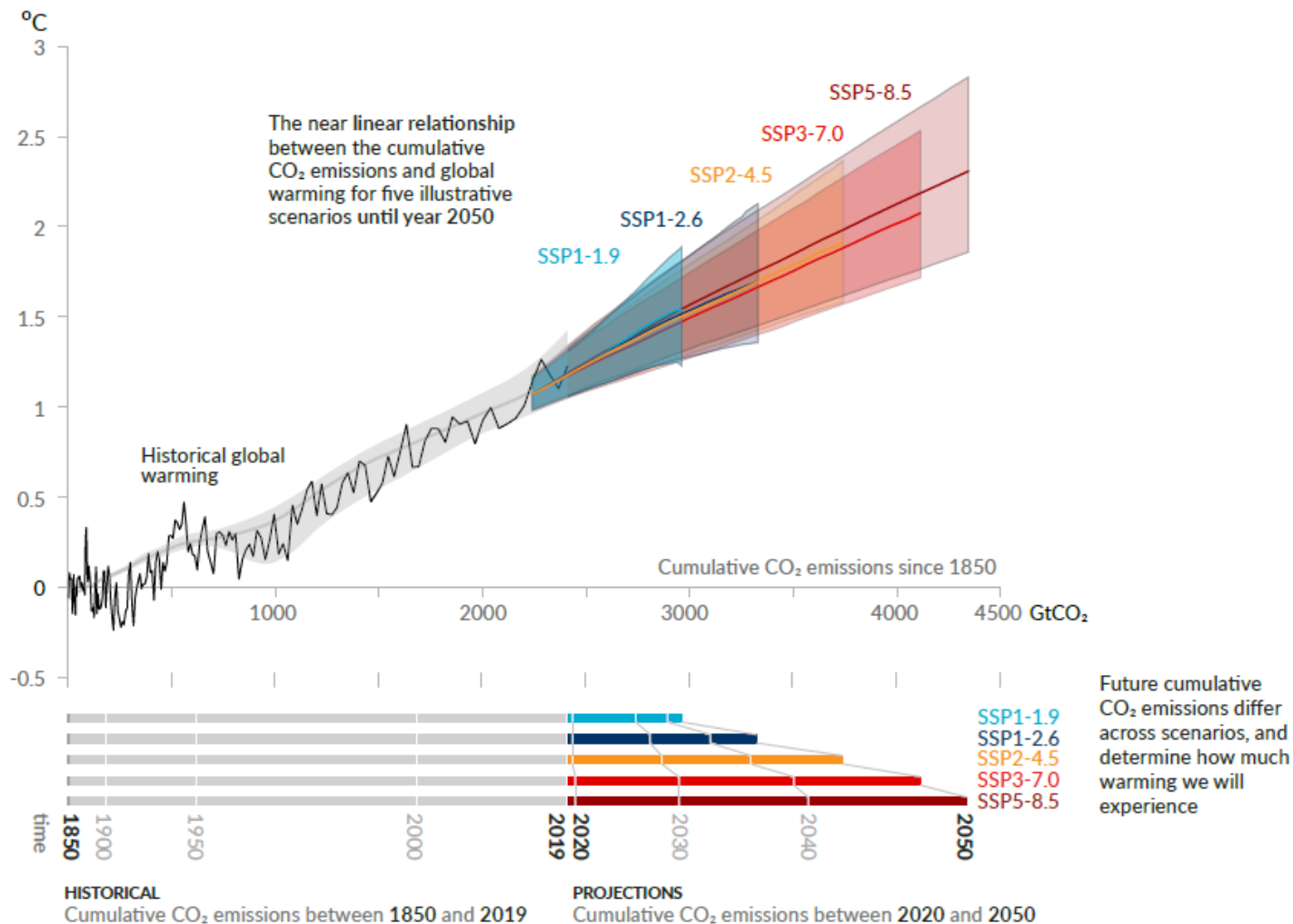
CO₂ concentration, ppmv



Temperature change from present, °C



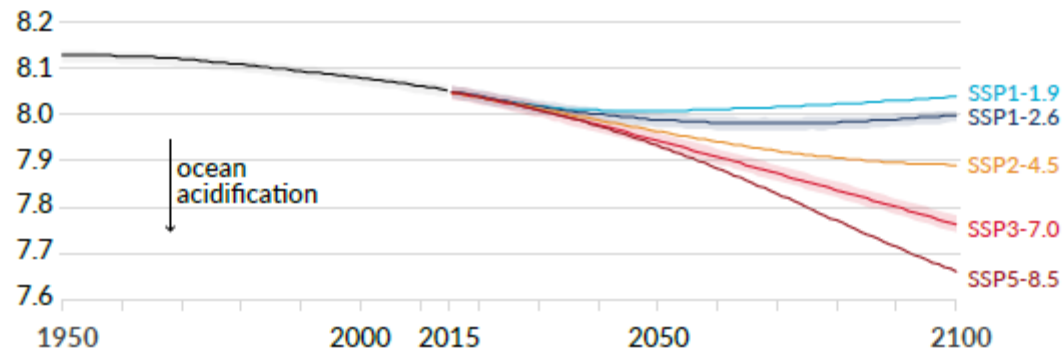
Global surface temperature increase since 1850-1900 ($^{\circ}\text{C}$) as a function of cumulative CO_2 emissions (GtCO_2)



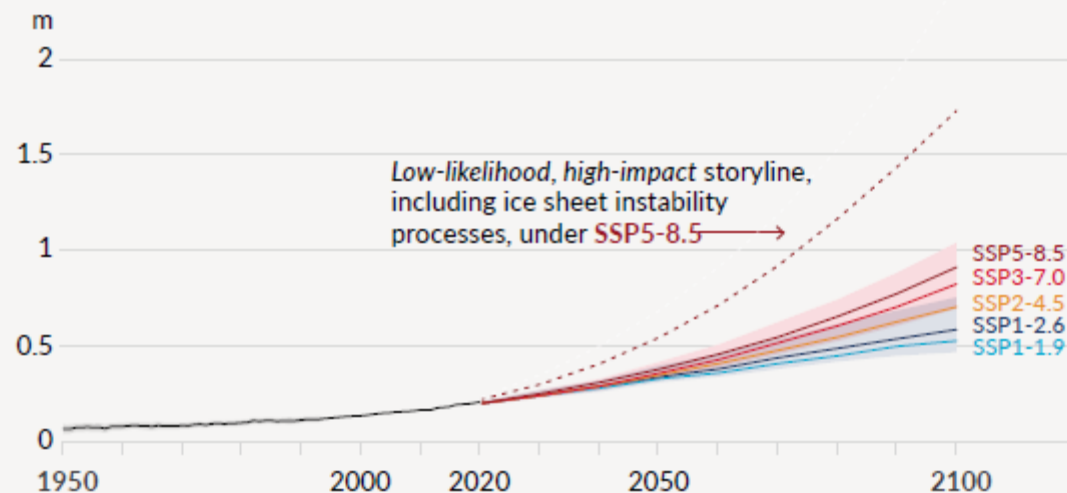
- SSPx refers to the Shared Socio-economic Pathway (SSP) describing the socio-economic trends underlying the scenarios 1-5.
- Source: <https://www.ipcc.ch/report/ar6/wg1/>

Surface Ocean CO₂ and pH

c) Global ocean surface pH (a measure of acidity)

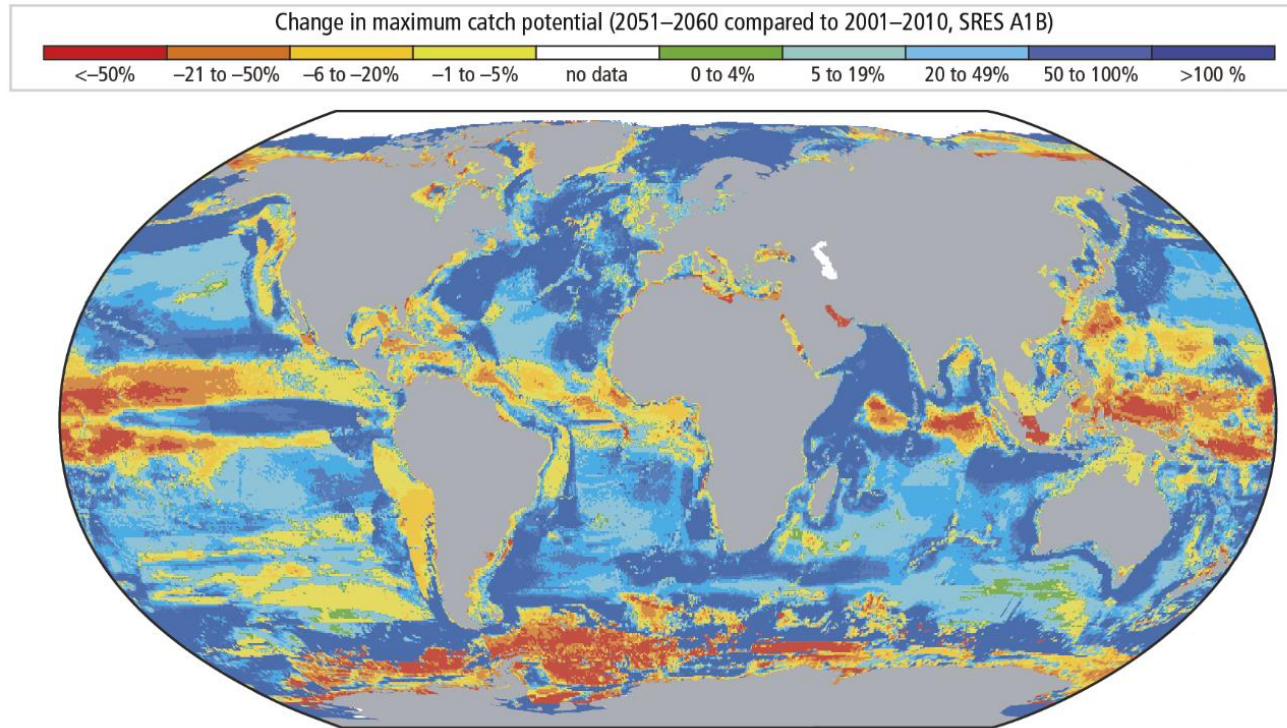


d) Global mean sea level change relative to 1900

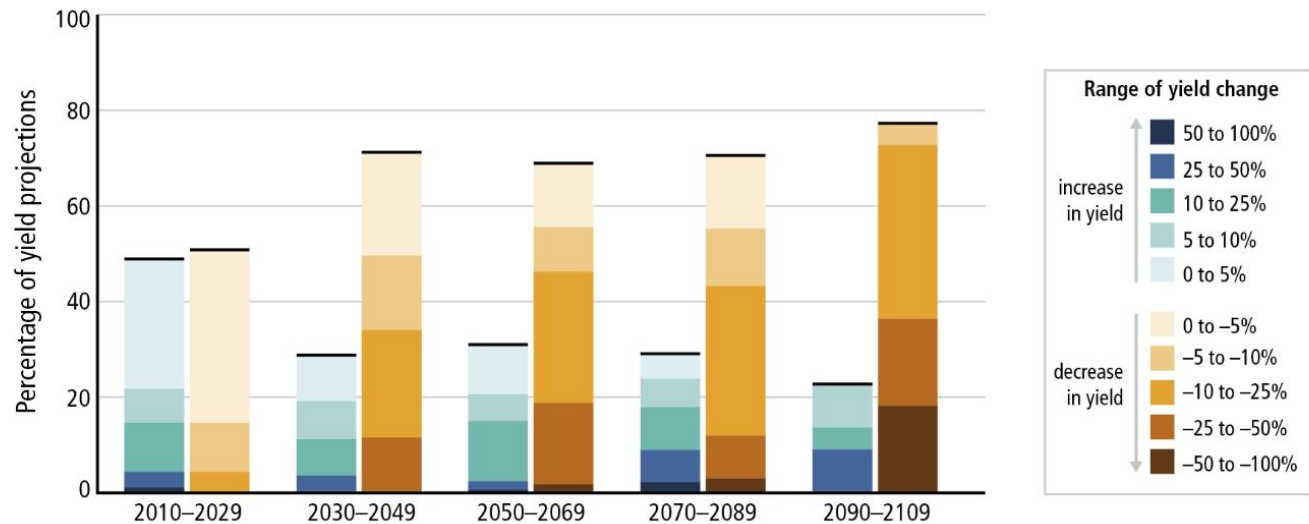


Climate change poses risks for food production

(a)

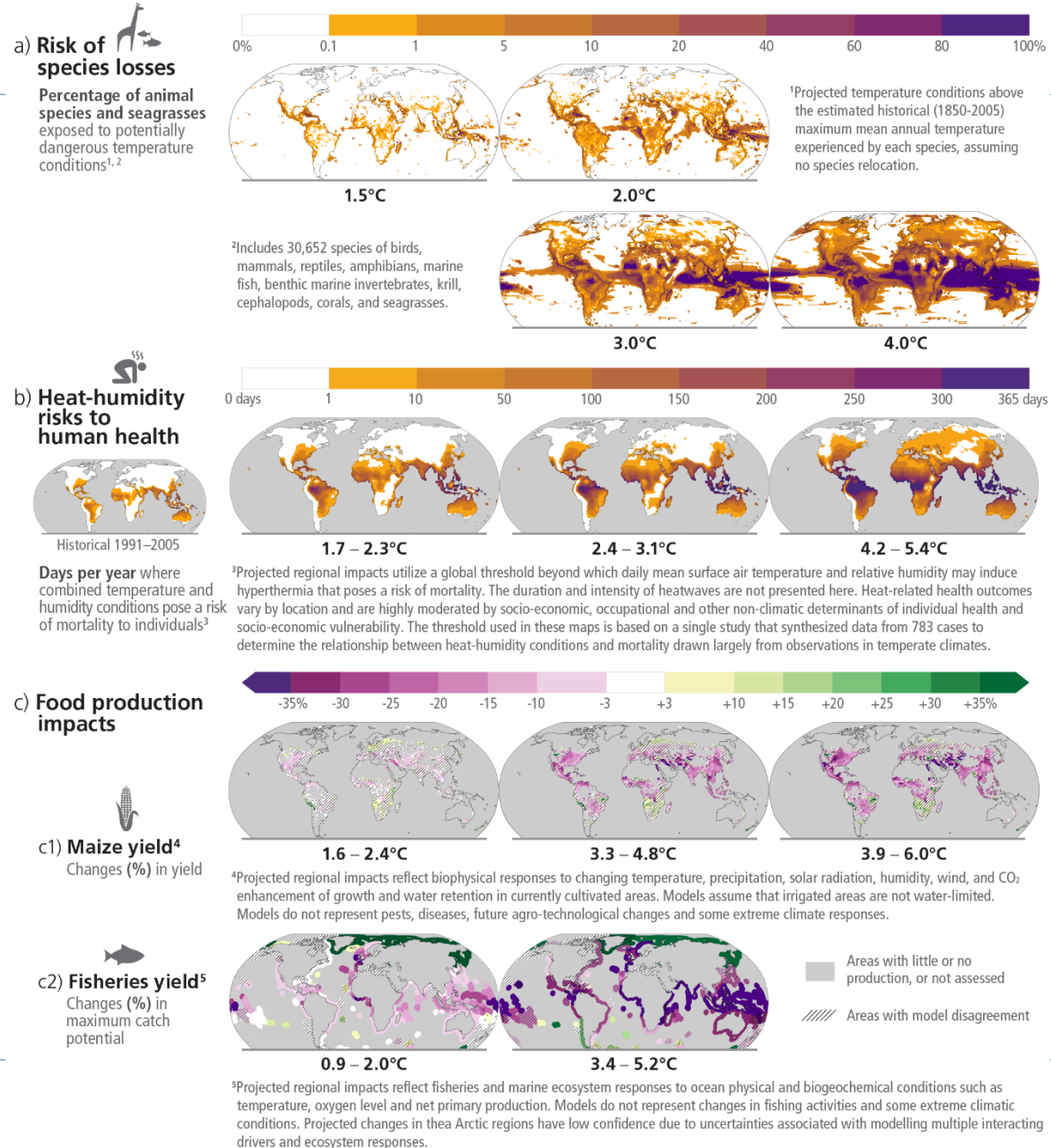


(b)



Future climate change is projected to increase the severity of impacts across natural and human systems and will increase regional differences

Examples of impacts without additional adaptation



Impacts of Climate Change

- ▶ Higher annual rainfall variation in low-latitude regions;
- ▶ More extreme weather conditions are recorded;
- ▶ Flooding, drought, uncontrollable hill fires, typhoons;
- ▶ Biodiversity seriously harmed; decrease in production of crops, lost of lives;
- ▶ Species extinction -> significant change in ecosystem;
- ▶ More serious infectious diseases -> higher burden on health services;
- ▶ Urban infrastructure (e.g. transportation, water, sanitation and energy systems) compromised by extreme and slow-onset event;
- ▶ More details at: [IPCC Sixth Assessment Report \(AR6\)](#).

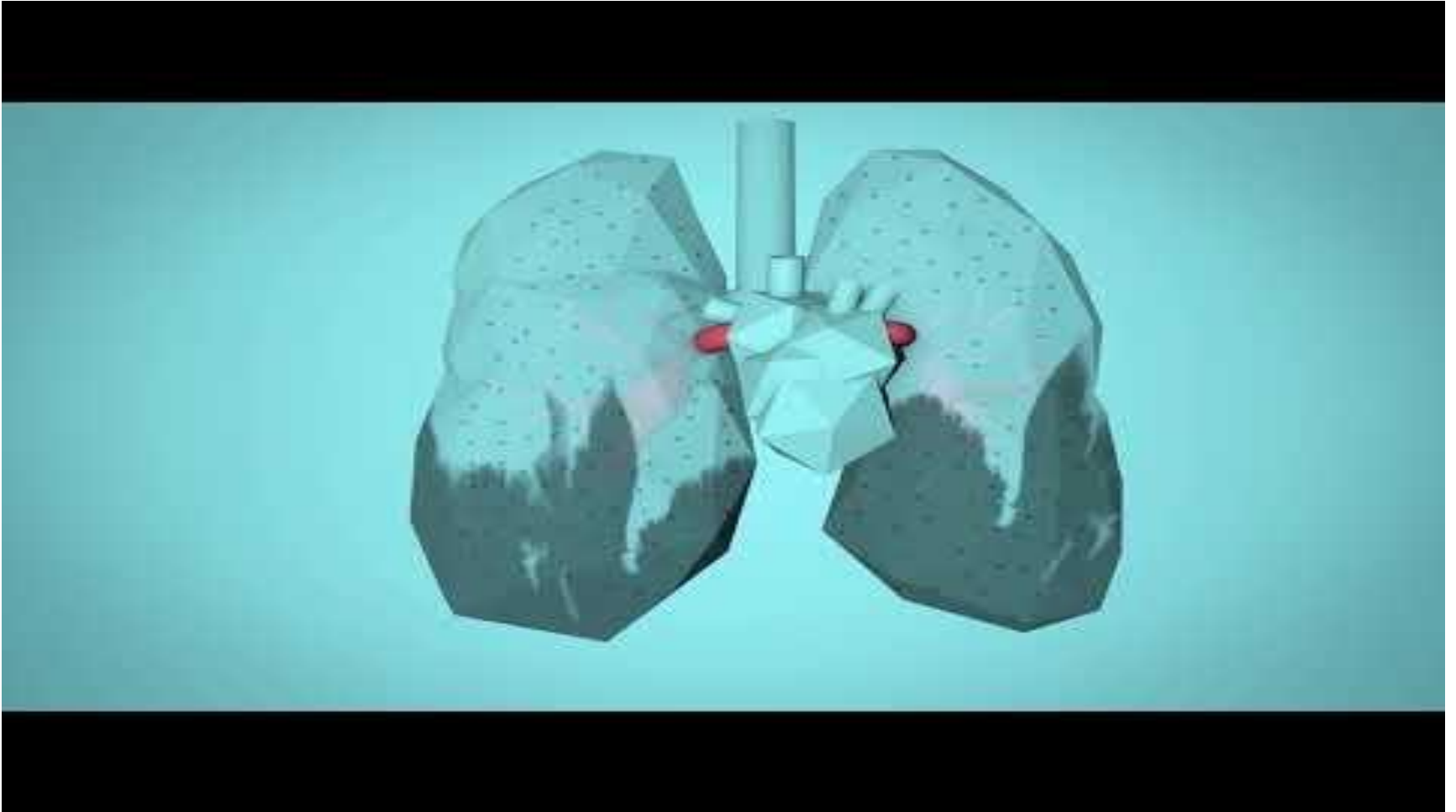
Harmfulness of air pollution

- ▶ According to the World Health Organization (WHO) report:
 - ▶ Air pollution can cause cancer (17th Oct., 2013)
<http://www.theguardian.com/environment/2013/oct/17/air-pollution-causes-cancer-world-health-organisation-who>
 - ▶ Outdoor air pollution has been officially classified as carcinogenic
 - ▶ the risk of developing lung cancer is **significantly increased** in people exposed to air pollution



Smog in Hong Kong
(source of image: The Guardian)

Air pollution impacts the body



<https://youtu.be/GVBeY1jSG9Y>

Main sources of air pollutants

- ▶ Burning of fossil fuels produce GHG:
 - ▶ Carbon dioxide CO₂
 - ▶ Methane CH₄
- ▶ and other pollutants including:
 - ▶ Carbon Monoxide, CO;
 - ▶ Sulphur oxides (SO_x);
 - ▶ Nitrogen oxides (NO_x); AND
- ▶ Respirable Suspended Particulates (RSP) - with diameter $\leq 10\mu\text{m}$ so named as PM₁₀; and Fine Suspended Particulates (FSP) - with diameter $\leq 2.5\mu\text{m}$ so named as PM_{2.5}.



Renewable Energy (RE) Topologies



Definition of Renewable Energy (RE)

- ▶ The [International Energy Agency](#) (IEA) defines renewable energy resources as those “derived from natural processes” and “replenished at a faster rate than they are consumed” (IEA and Eurostat, 2005*).
- ▶ The IEA definition of RE includes: “electricity and heat derived from solar, wind, ocean, hydropower, biomass, geothermal resources, and biofuels and hydrogen derived from renewable resources”.
- ▶ Renewable energy does not include energy resources derived from fossil fuels, waste products from fossil sources or waste products from inorganic sources. (source: [Texas Renewable Energy Industries Alliance](#)).

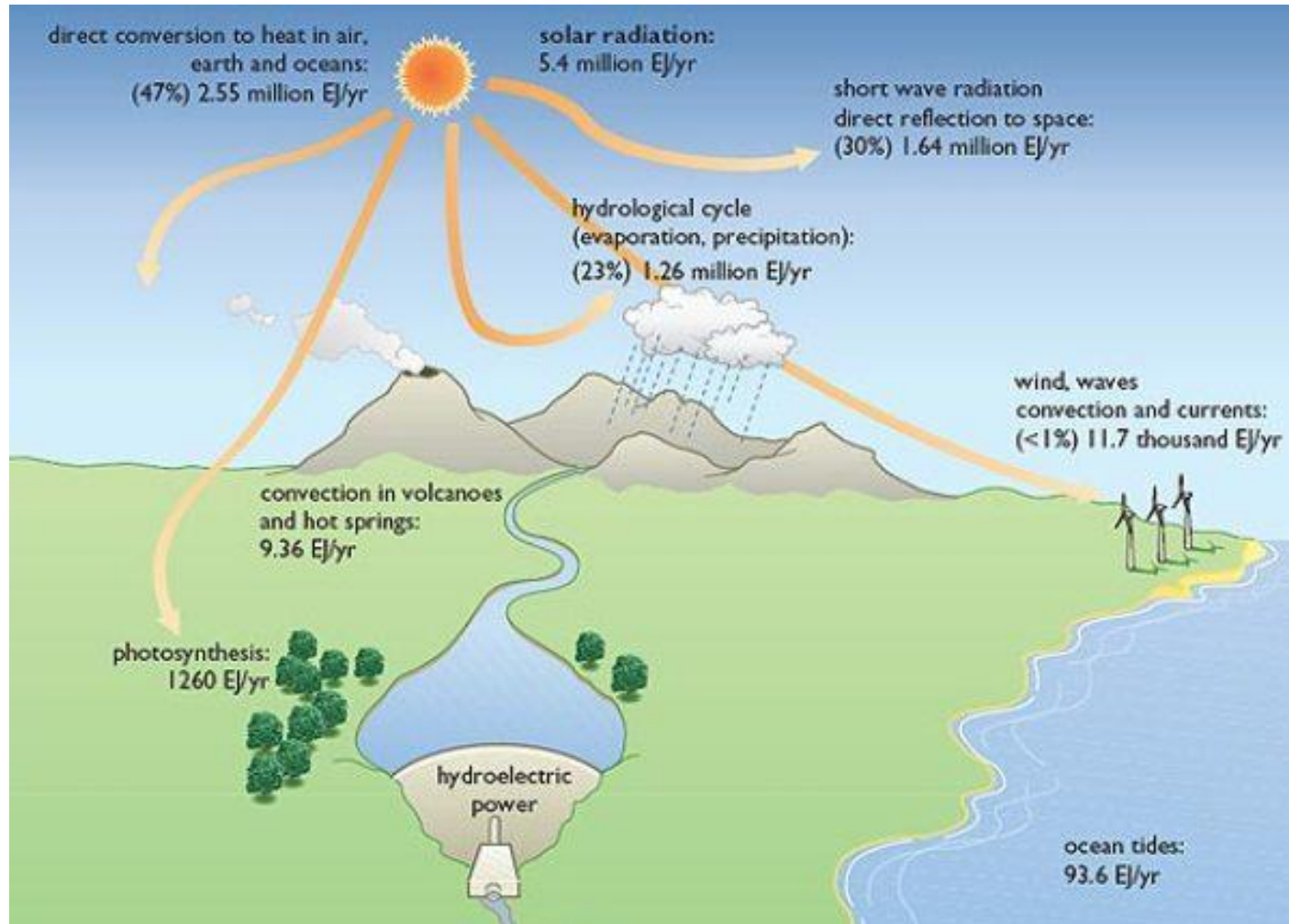
** IEA, OECD, EUROSTAT, 2005. “Energy Statistics Manual, International Energy Agency, Organisation for Economic Co-Operation and Development and Eurostat”, Paris, France.*

Classification of energy sources

Primary energy		Converted by	Secondary energy (energy carrier)
Non-renewables	Crude oil	Oil refinery	Fuel oil, Gasoline, Diesel
	Coal	Fossil fuel power station	Enthalpy (heat), mechanical work or electricity
	Natural gas		
	Natural uranium	Nuclear power plant	Electricity
Renewables	Solar	Photovoltaic power plant	Enthalpy or electricity
		Solar thermal plant	
	Wind	Wind turbine / farm	Mechanical work or electricity
	Falling / flowing water	Hydropower plant	
		Tidal power station	
		Wave power station	
	Biomass	Biomass power station	Enthalpy or fuel or electricity
	Geothermal	Geothermal power station	Enthalpy or electricity

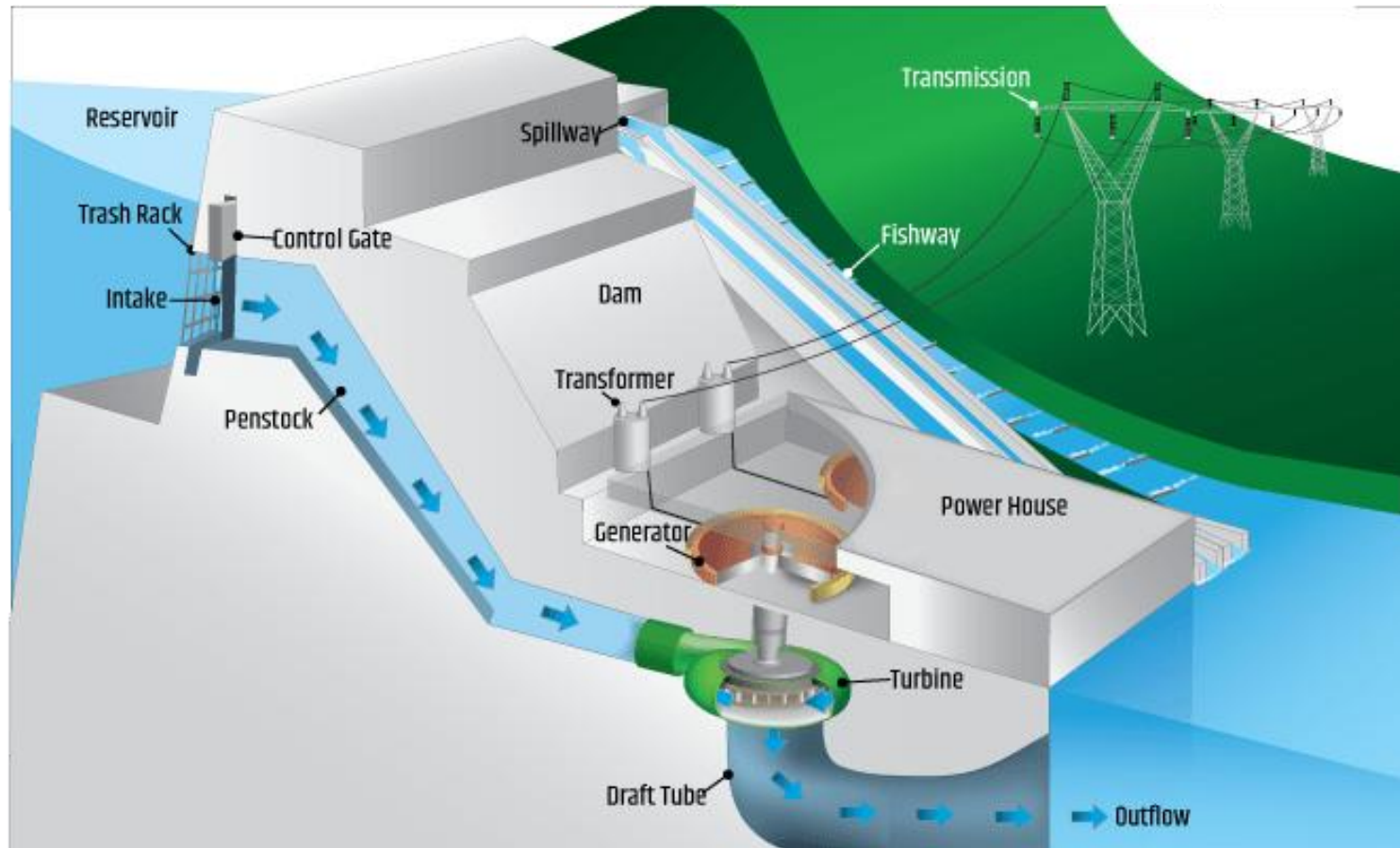


Sources of Renewable Energy



Note: 1EJ = 1 exajoule = 1×10^{18} J

Hydropower



Source of image: <https://www.energy.gov/eere/water/types-hydropower-plants>

Hydropower

- ▶ The driving force for hydropower is the hydrologic cycle.
- ▶ In this cycle, the sun's energy is absorbed by surface water and soil at low elevations, causing water to evaporate to form water vapour. Water vapour also is produced from transpiration of plants. Clouds form from this water vapour. The vapour later condenses to form water droplets, which will eventually fall as precipitation. If this precipitation lands on areas of high elevation, the water will have relatively high potential energy as gravity causes this water to run off in streams.
- ▶ The water's potential energy is converted into kinetic energy, frictional heating, and mechanical energy. Hydropower technologies attempt to convert some of this energy into mechanical or electrical energy.

Hydropower

- ▶ To determine the power available in a stream or river with Bernoulli's equation:
 - ▶ $P_m = \eta \rho g Q H \dots \dots \dots (1.1)$
where:
 - ▶ P_m is the mechanical power produced at the turbine shaft (Watts),
 - ▶ η = the hydraulic efficiency of the turbine, ρ is the density of water (1000 kgm^{-3}),
 - ▶ g = the acceleration due to gravity (9.81 ms^{-2}),
 - ▶ Q = the volume flow rate passing through the turbine (m^3s^{-1}),
 - ▶ H = the **effective pressure head** of water across the turbine (m).
- ▶ The best turbines can have hydraulic efficiencies in the range 80 to over 90%, although this will reduce with size. Micro-hydro systems (<100kW) tend to be 60 to 80% efficient.

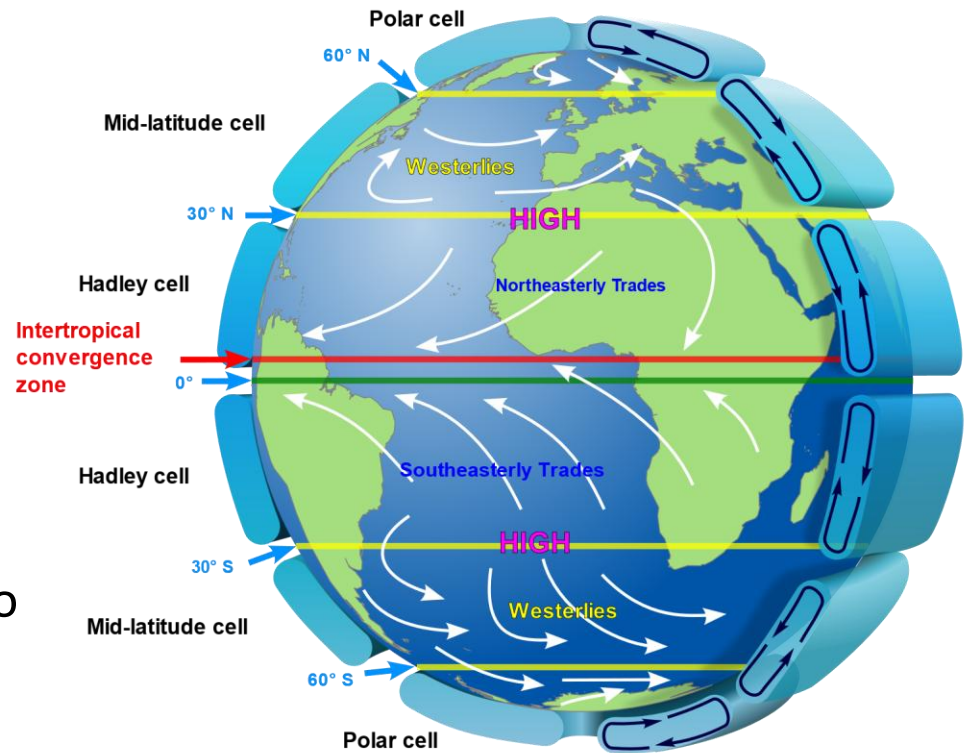
Wind power



Shantou, Guangdong, China

Wind power

- ▶ Wind is derived indirectly from solar energy.
- ▶ Wind is generated by the movement of air from regions of high pressure to regions of low pressure, with these differential pressures established by thermal gradients.
- ▶ While wind directions may seem to change with frequency in most places there are prevailing wind patterns that make it easier to predict and harness the power of wind.



Source of image:

https://en.wikipedia.org/wiki/Atmospheric_circulation

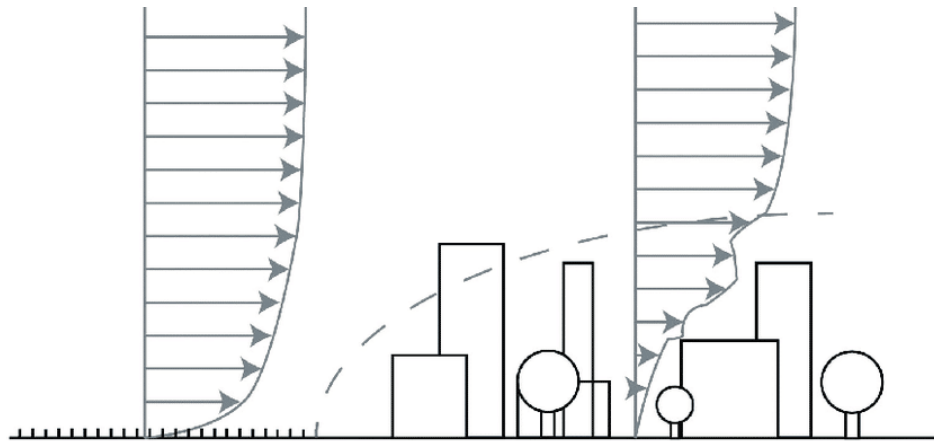
Wind turbine

- ▶ Wind turbine uses wind to rotate the blades which drives the gearbox behind it to increase the rotational speed to generate electricity.
- ▶ Current wind turbine technology can start generating electricity above 3ms^{-1} of wind speed.
- ▶ Current large-scale wind turbines have peak power of $>2000\text{kW}$ located on land, along coastline or offshore.
- ▶ Latest generation of wind turbines have peak power at MW scales and is increasing.

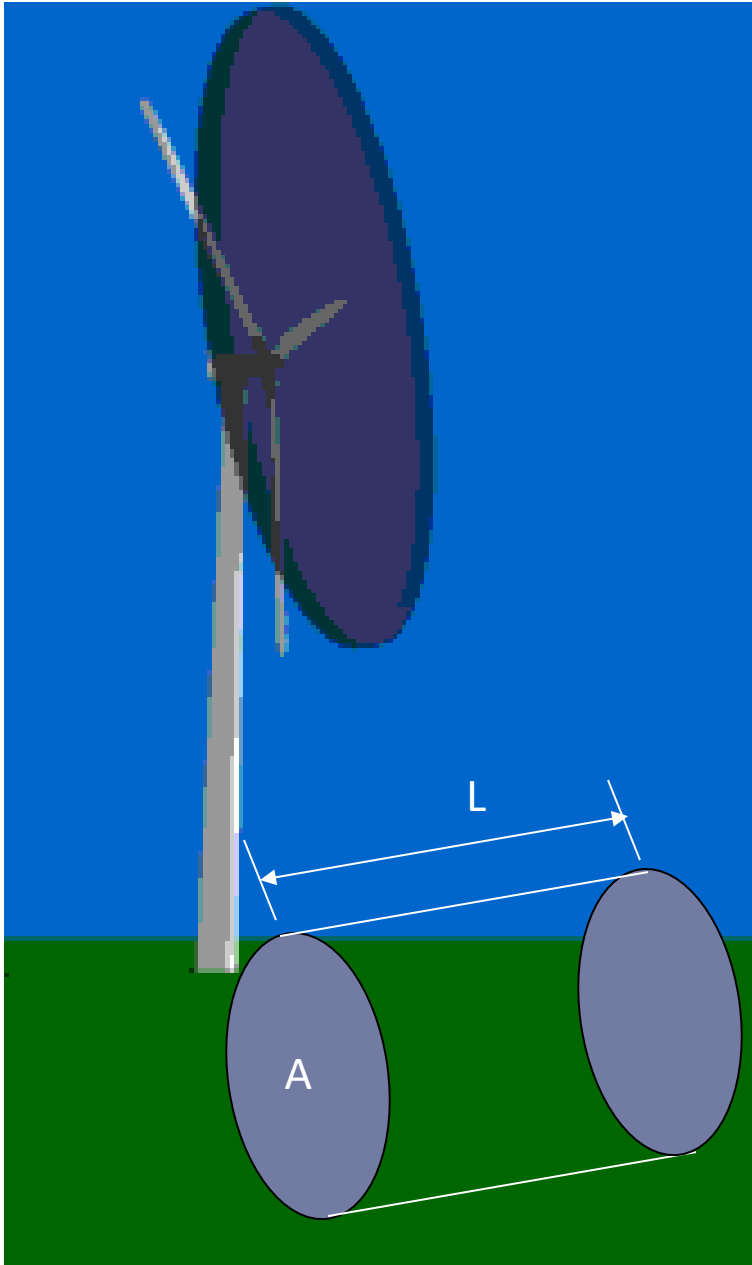


Local air circulation

- ▶ With increasing proximity to the surface of the earth, wind is increasingly influenced by local effects.
- ▶ Due to friction with the (rough) surface of the earth, the geostrophic wind (air circulation due to local effects) is reduced and almost standstill at ground level.



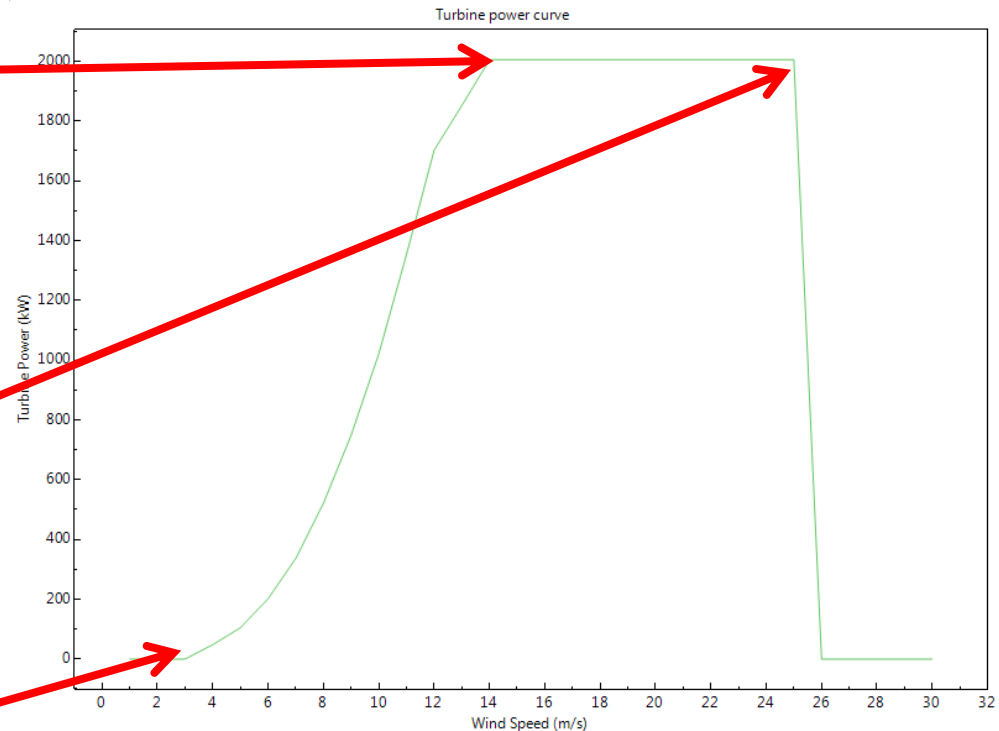
Source of image: <https://www.mecaenterprises.com/getting-high-on-wind/>



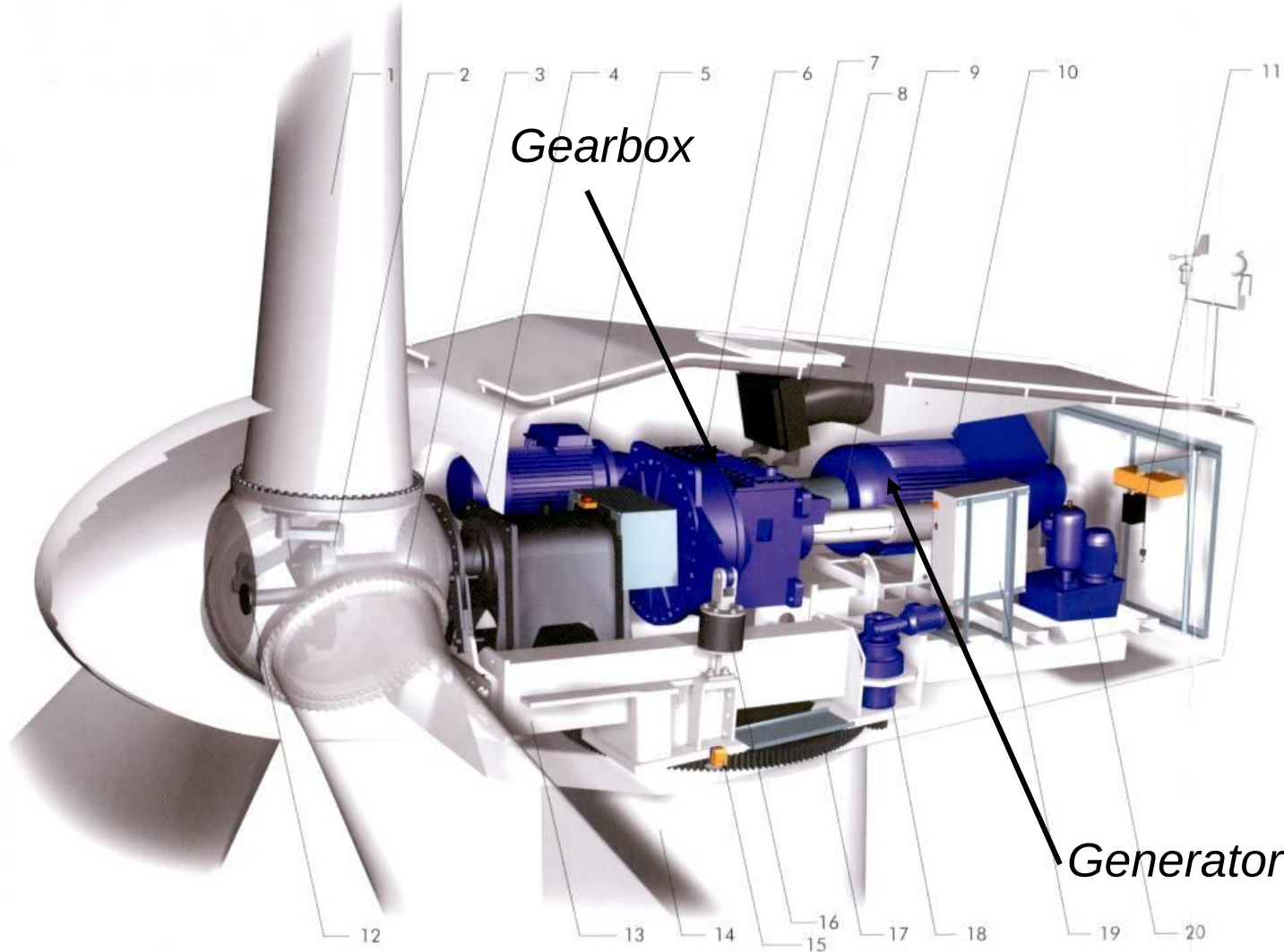
- ▶ K.E. = $\frac{1}{2} * \text{mass} * \text{velocity}^2$
- ▶ Wind volume = $A * L$, wind power is:
- ▶ $P_w = d(K.E.) / dt$
 $= d(\frac{1}{2} m v^2) / dt$
 $= d(\frac{1}{2} \rho A * L * v^2) / dt$
 $= \frac{1}{2} \rho A v^2 dL / dt$
 $= \frac{1}{2} \rho A v^3 \quad (1.2)$
- ρ = density of air
- A = area of wind blades
- ▶ *Wind power is proportional to cube of wind speed!*

Wind turbine power curve

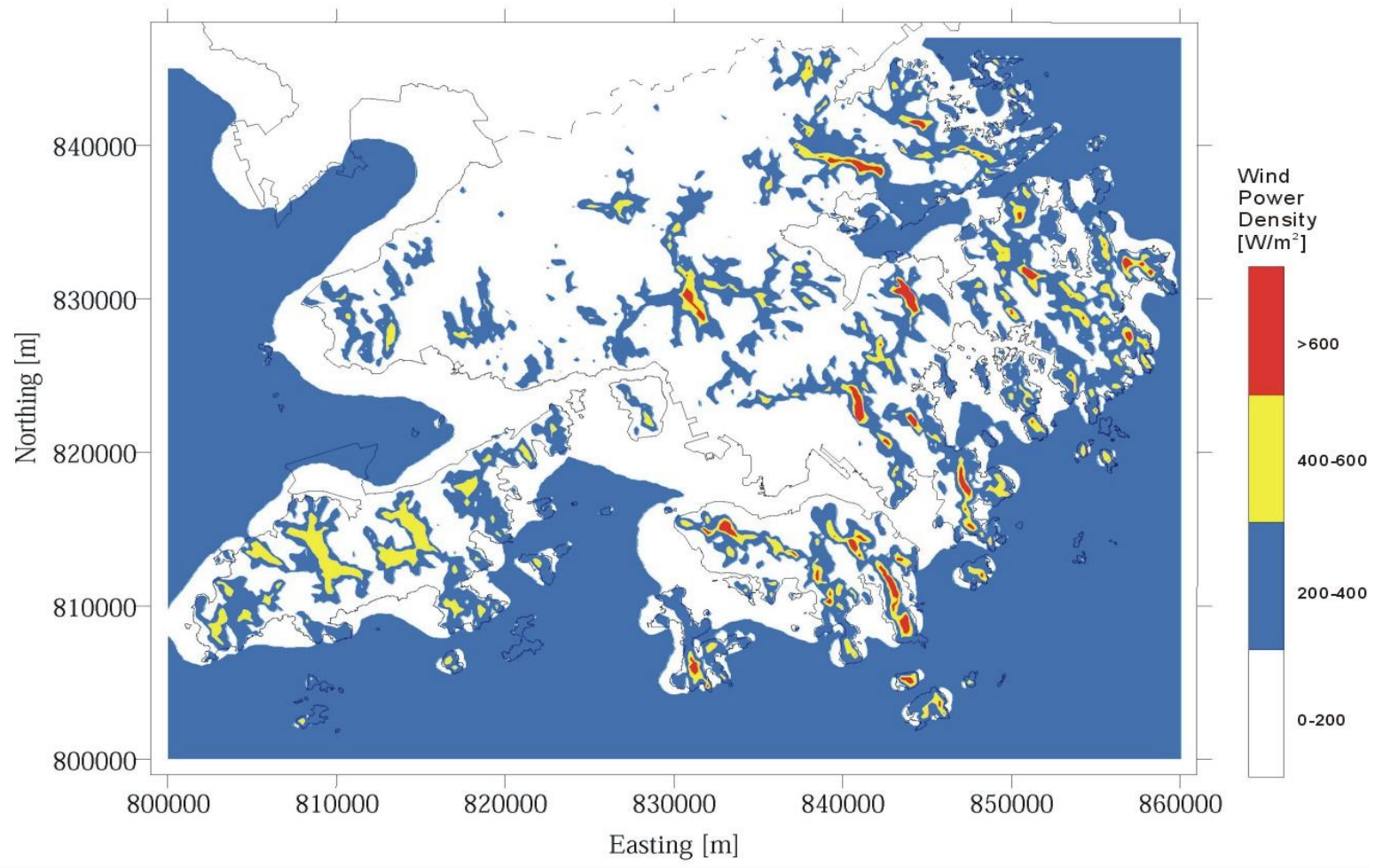
- $P_W \propto v^3$
- Rated output -> maximum power output
- Protecting the wind turbine from being damaged at high wind speed -> cut-out wind speed
- Need to overcome frictions before generation -> cut-in wind speed



Components inside a wind turbine

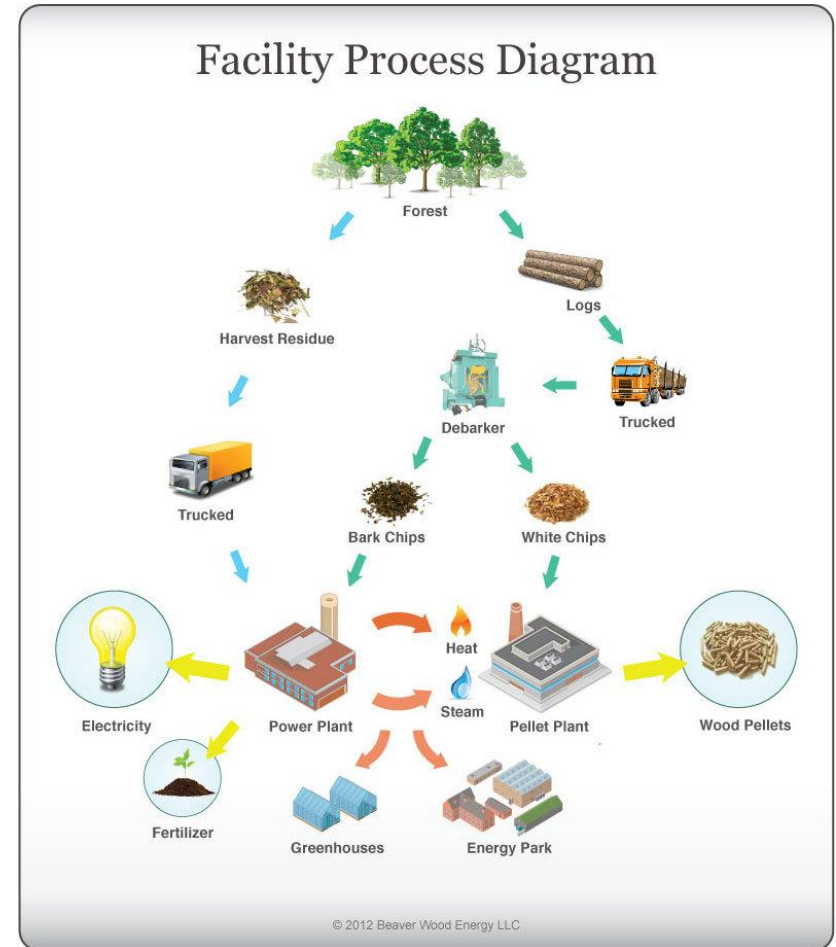


Wind resources of Hong Kong



Biomass

- ▶ It's a process of large scale of growing plants then burning or conversion process to produce energy. There are six main ways to process biomass:
 1. direct combustion (for power);
 2. anaerobic digestion (for methane-rich gas);
 3. fermentation (of sugars for alcohols);
 4. oil exaction (for biodiesel);
 5. pyrolysis (for biochar, gas and oils);
 6. gasification (for carbon monoxide and hydrogen-rich syngas).

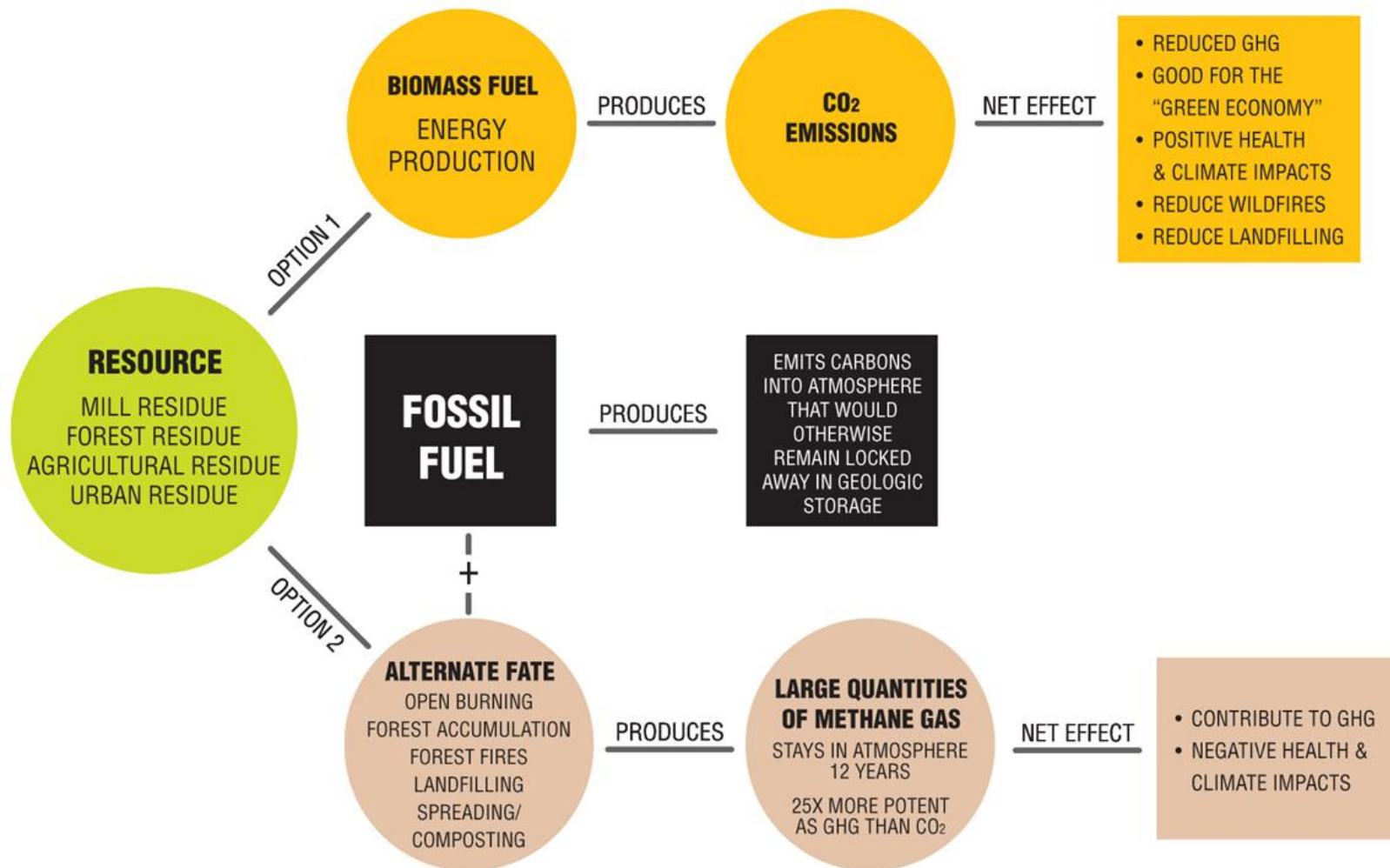


Source of image: Beaver Wood Energy, USA

Biomass

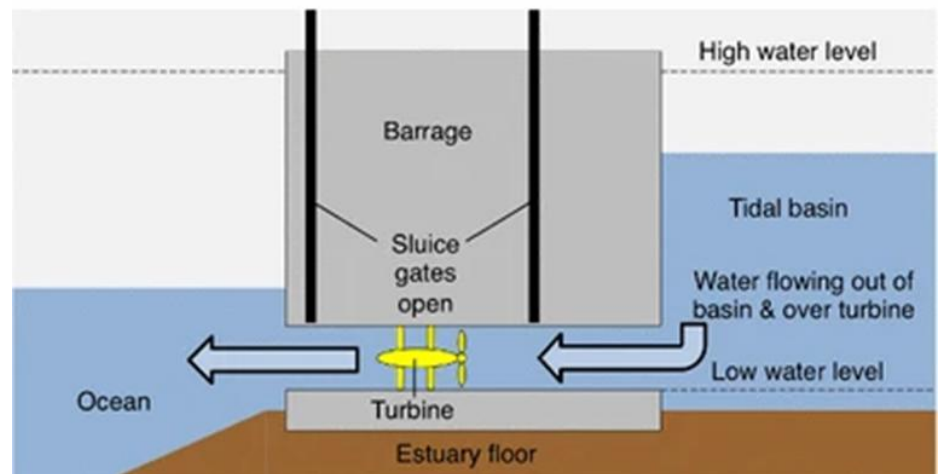
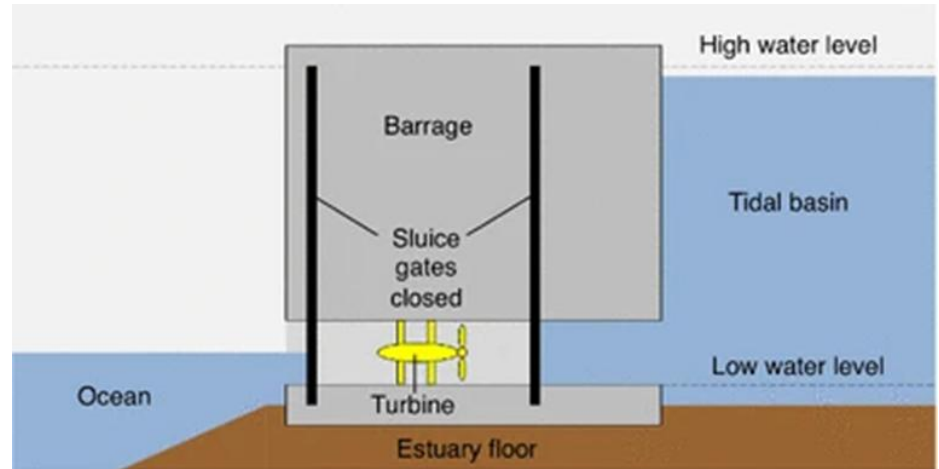
- ▶ Throughout the process net release of CO_2 is zero theoretically, therefore it is also a type of RE.
- ▶ Biofuels are the combustion of processed materials like straw, forestry waste and wood chips which are used directly to produce electricity.
- ▶ Biogas, produced by anaerobic digestion. Main components of biogas include methane and CO_2 . This process happens in places such as marshes and digestive system of animals.
- ▶ Biomass has long history of utilisation. Wood was the most widely used form of energy before the discovery of oil. Recent technologies enable conversion of plants to methanol / ethanol directly or by catalytic conversion.

Biomass net negative GHG impact



Tidal power

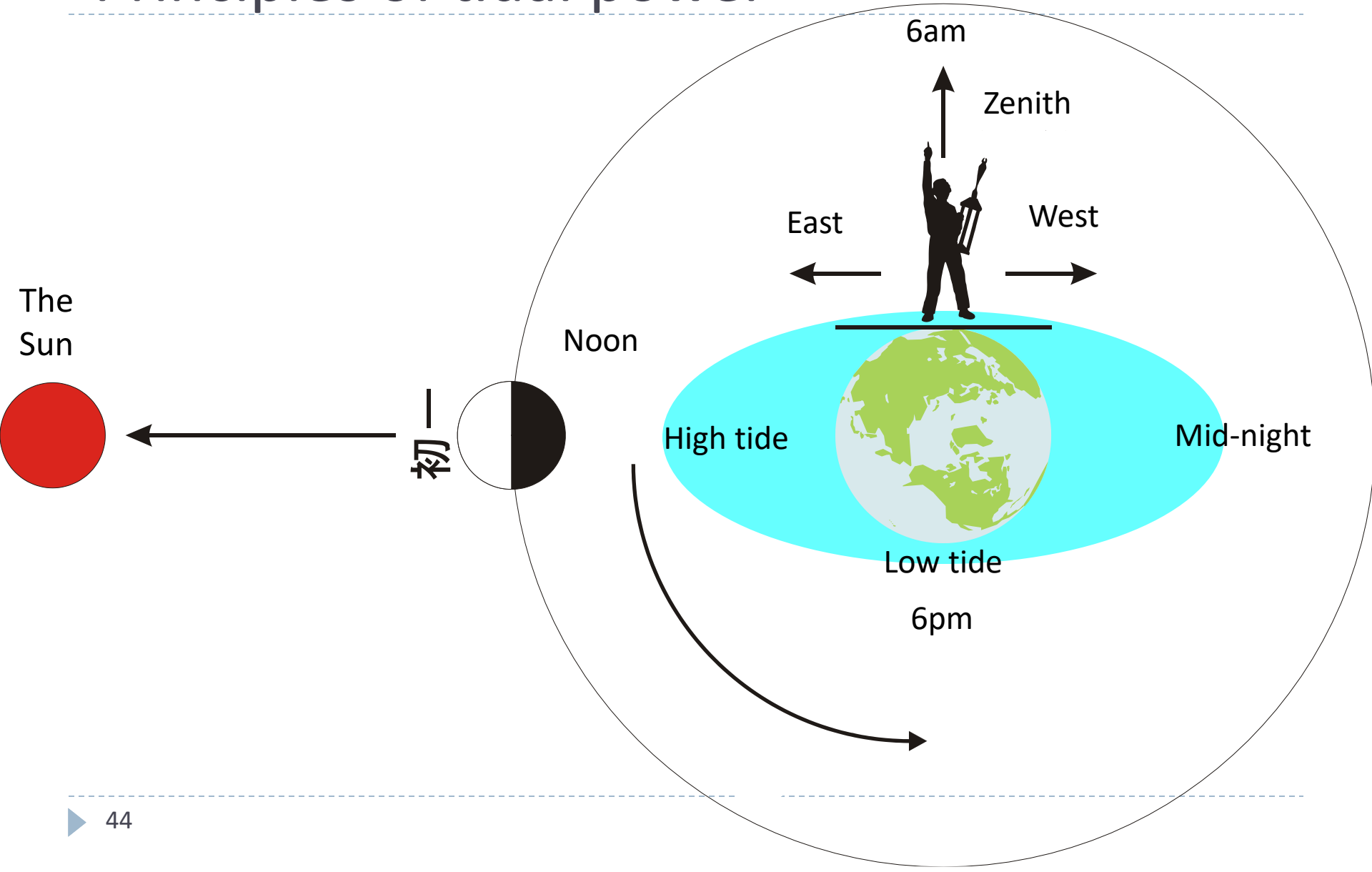
- ▶ Tidal power makes use of the interaction of solar and lunar gravitational attraction which produces high and low tides twice a day.
- ▶ Works by building Barrage to contain water after high tide, then water has to pass through a turbine to return to low tide.
- ▶ In general, first and fifteenth day of lunar calendar have highest potential difference when the Sun, the Earth and the Moon align. Potential difference also depends on location.



Source:

<https://www.wyretidalenergy.com/tidal-barrage/how-a-barrage-works>

Principles of tidal power



How tidal power works?



Barrage Tidal Power



Generation



Tidal power

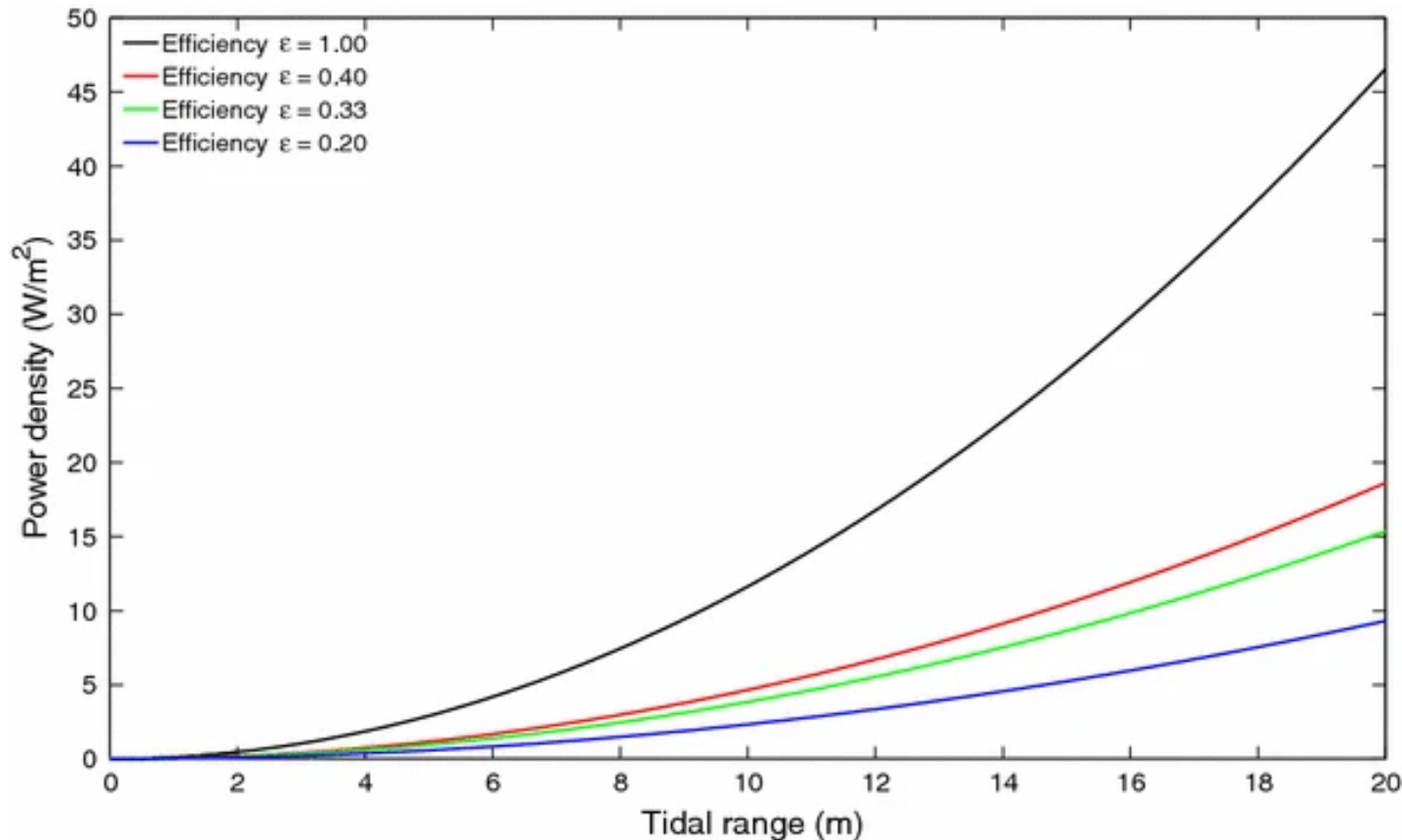
- ▶ Power density in terms of power per unit of basin area:

$$P_d = \frac{P}{A_b} = \frac{\rho g H^2}{2t} \varepsilon.$$

(1.3)

where H = Hydraulic head (m); and

ε = device efficiency



source: <https://link-springer-com.eproxy.lib.hku.hk/article/10.1007/s40722-016-0044-8>

Tidal power analysis

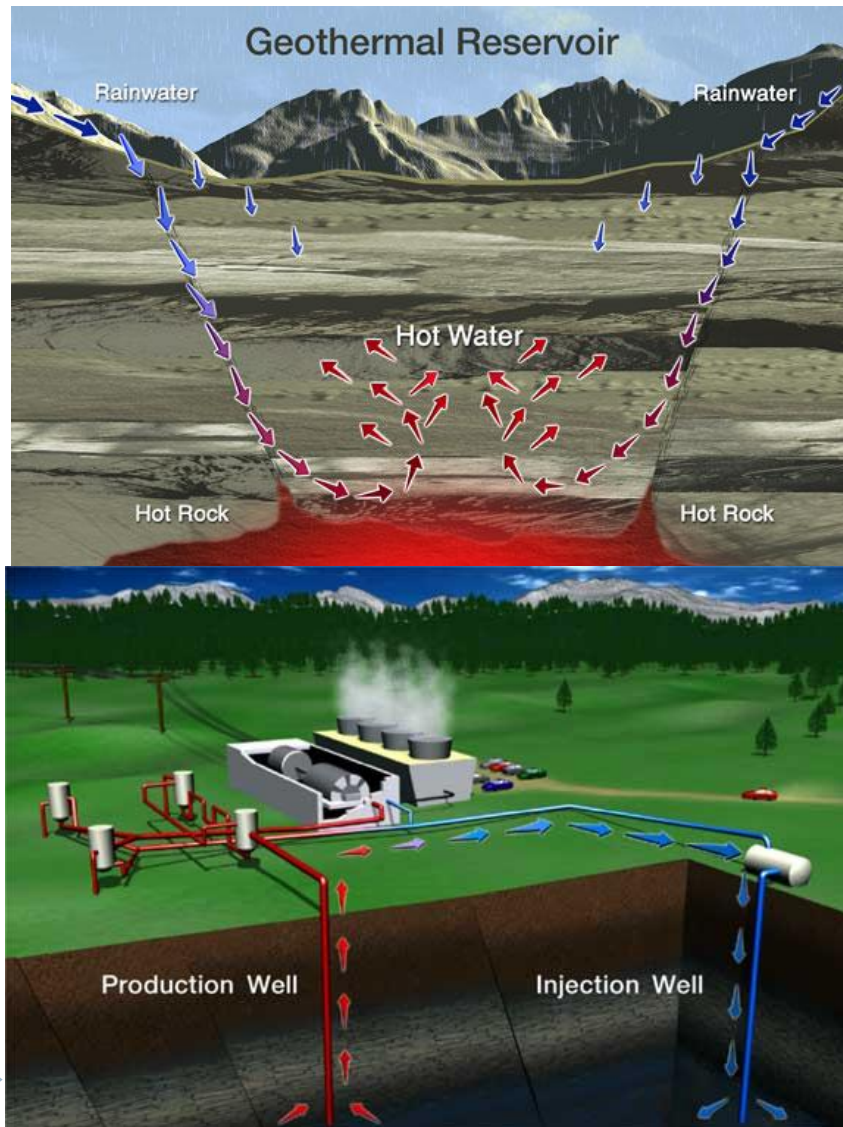
► Advantages of Tidal power:

1. tidal stream energy is the most reliable source of renewable energy since it is not affected by global climate change;
2. unaffected by weather: predictable tidal currents, almost invariant with weather;
3. no visual nor noise pollution problem;
4. water is about 400 times more powerful than wind so each turbine can be a lot smaller.

► Disadvantages of Tidal power:

1. possible effects on sea mammals affecting their navigation;
2. very location specific - dependent on the tidal range;
3. infant technology - the turbine technology use in sea water is still developing.

Geothermal energy



- ▶ Geothermal energy under the Earth's surface comes from hot rocks and liquids produced by recent volcanic activities. One of the principal sources of geothermal energy comes from large reservoirs of hot water trapped in the permeable rock formations below the surface.
- ▶ Only when the thermal energy is located at shallow enough depths that it can be recovered and put to use.
- ▶ Location dependent for large scale energy extraction.

Geothermal systems and resources

- ▶ Hot wells at the surface of the earth, and the volcanic areas demonstrate the geothermal energy potential.
- ▶ The geothermal heat flow is independent of daily or seasonal influences. Thus, there is no variation in the energy supply in human time dimensions.
- ▶ The migration of locally confined heat centres within the earth's crust can, for example, lead to a change in the location of the reservoir through tectonic processes. Also, the processes associated with the rising salts in the sediment basin (i.e. the growing of salt domes) also play a certain role. However, all these slow time-related changes in the heat content have no impact on energetic utilisation.

Geothermal energy systems analysis

► Advantages of Geothermal energy:

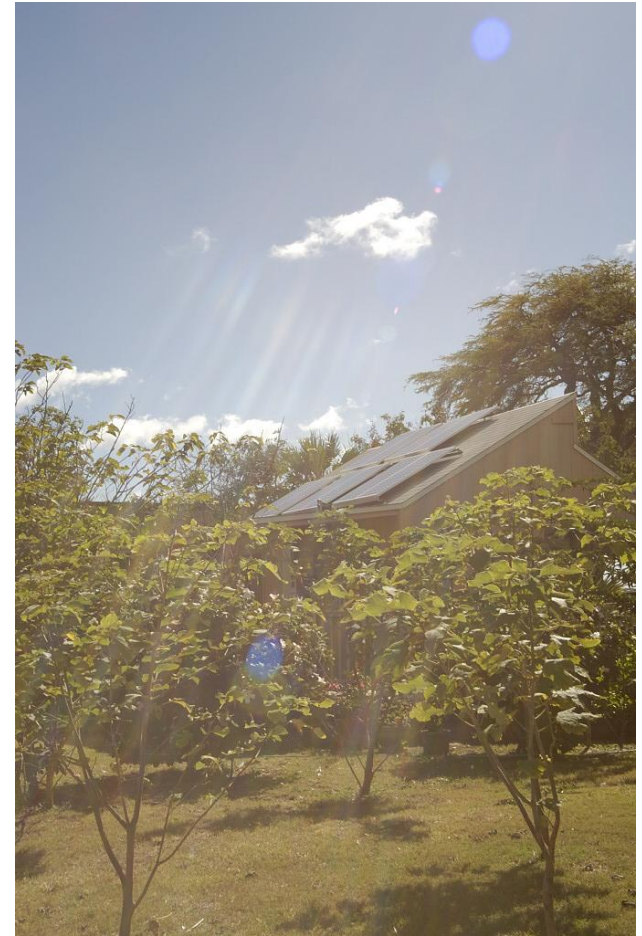
1. provides consistent energy output unlike other renewables (no intermittency);
2. helps stabilize electricity grids (offset the risks associated with an energy system based on variable supplies);
3. offers direct and efficient heating capabilities (in [building heating systems](#)).

► Disadvantages of Geothermal energy:

1. slower adoption rate compared to other renewables;
2. high initial investment costs for Geothermal power plants;
3. limited geological data availability;
4. shortage of skilled workforce.

Solar energy

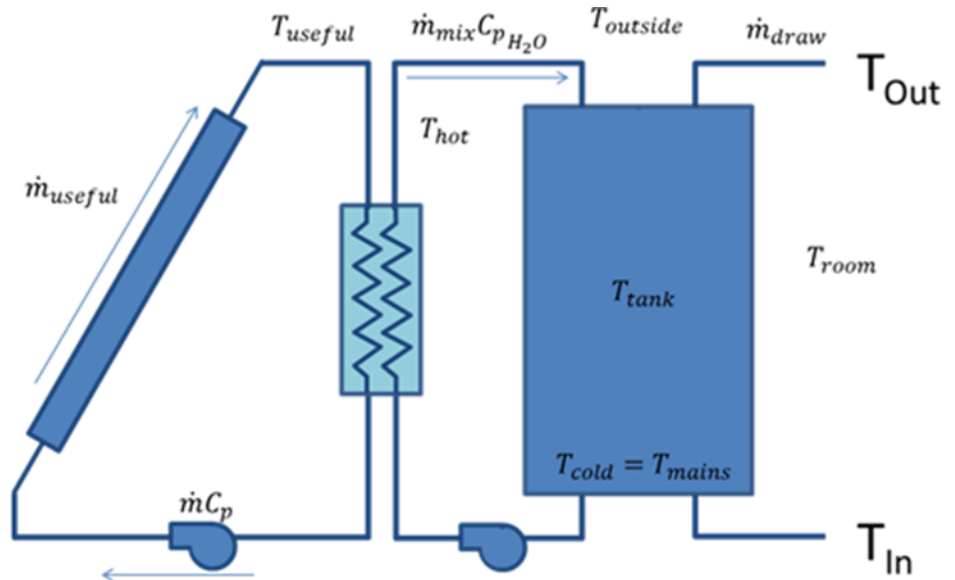
- ▶ The human race has a long history of harvesting solar energy.
- ▶ It comes to earth in the form of electromagnetic (EM) waves.
- ▶ Most types of renewable energy are derivatives of solar energy.
- ▶ Solar thermal systems absorb the heat from the sun to heat up a media of thermal storage for further use.
- ▶ Light-using systems include photovoltaic and daylighting to light built environment naturally.



Solar thermal systems

- ▶ This is the process of extracting the *heat energy* directly from the sunlight.
- ▶ Relatively low cost, but the utilisation of the heat is another issue.
- ▶ Sometimes the heat is used for heating up hot water directly (most common form of solar thermal system) for domestic use.
- ▶ Sometimes the heat is concentrated and stored for boiling of water, which produces steam to provide energy input for engine to generate electricity.
- ▶ Another way to use the heat could be used in heat pump for air-conditioning – but very expensive equipment and control.

Solar thermal for domestic hot water



- ▶ Determining required capacity: $Q_{hw} = V_{hc} \times C_w \times D_q$ (1.4)
 - ▶ where Q_{hw} = Daily heat requirement (MJ)
 - ▶ V_{hc} = Average daily hot water consumption (m^3)
 - ▶ C_w = Specific volumetric heat capacity of water ($=4.2 \text{ MJ}/m^3/^\circ\text{C}$)
 - ▶ D_q = Temperature difference between incoming water and the required hot water temperature ($^\circ\text{C}$)

Concentrated Solar Power (CSP)

- The main difference between these four technologies is the way they concentrate sun light.



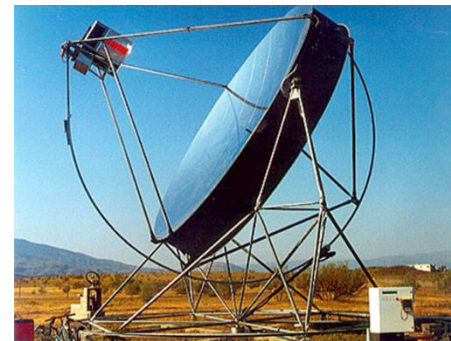
Solar Tower



Parabolic Trough

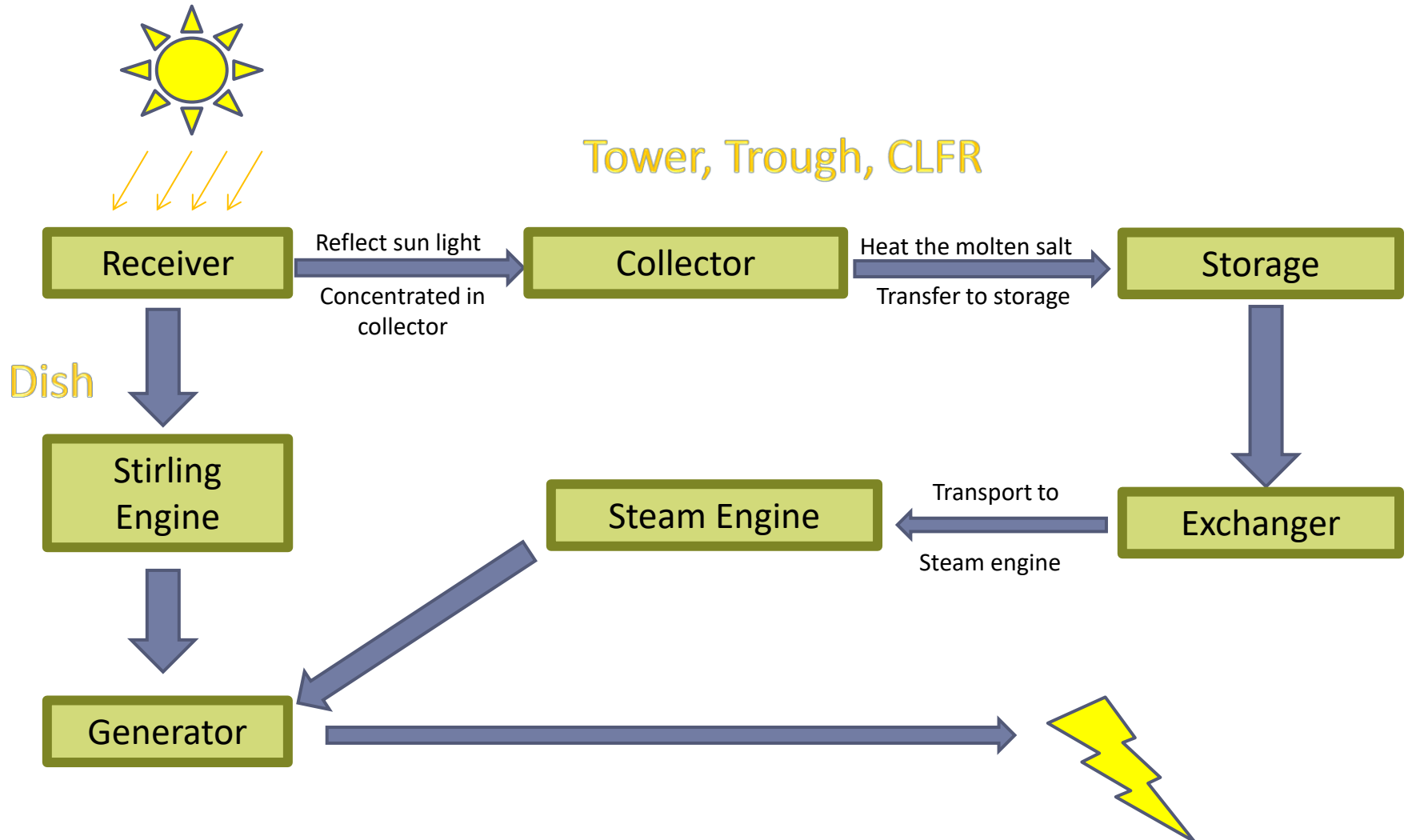


Compact Linear Fresnel Reflectors



Parabolic Dish

Operating principles of CSP

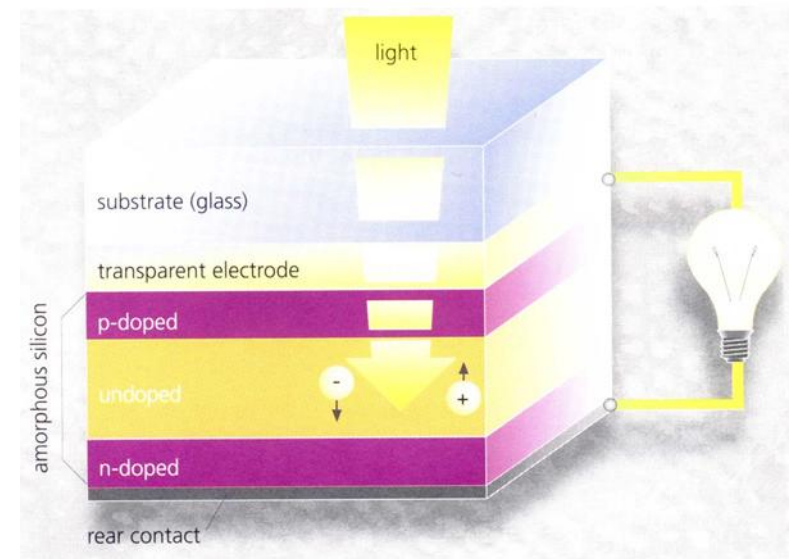


Operating principles of CSP

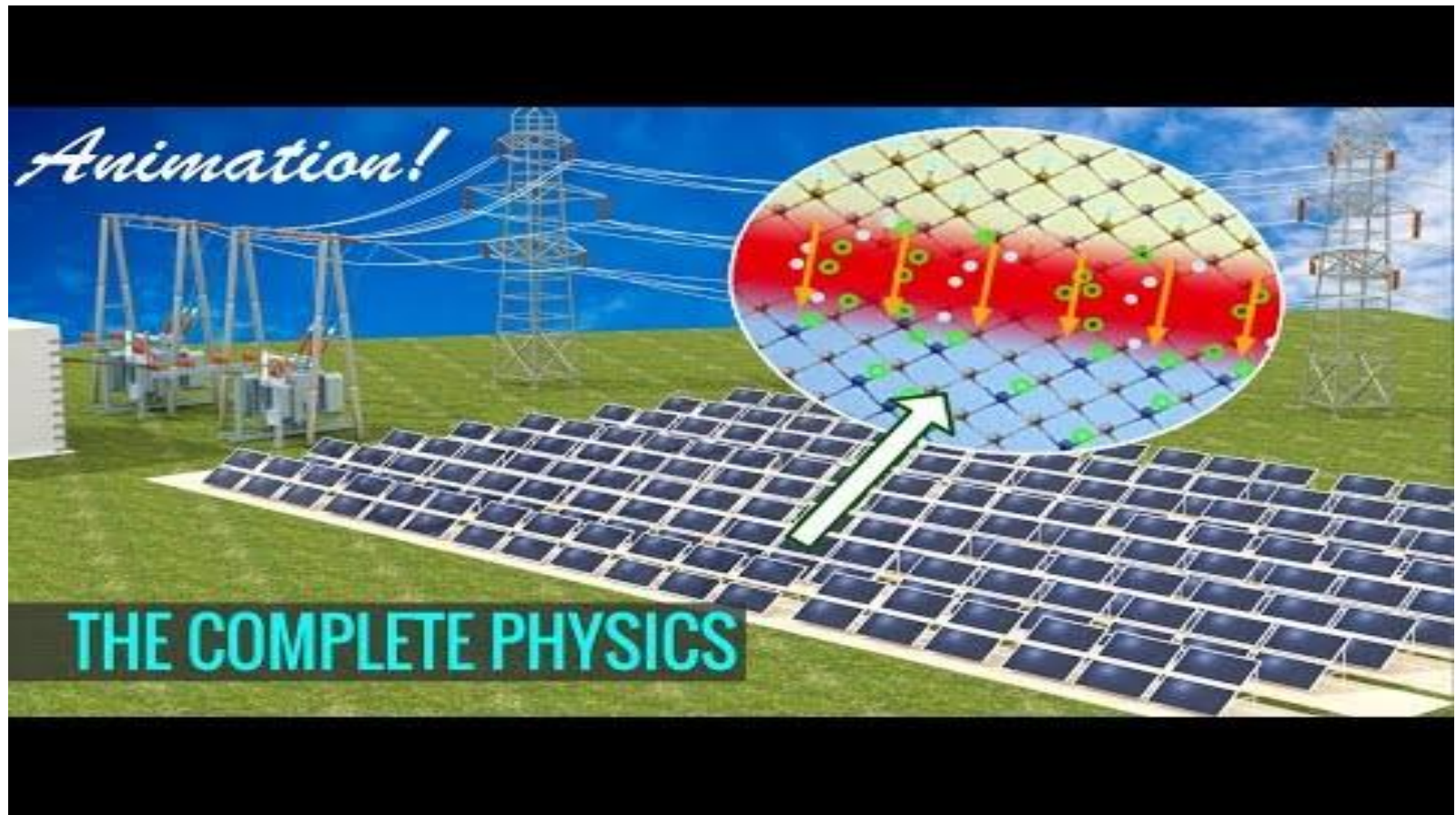
- ▶ After collecting solar thermal energy, Parabolic Dish directly uses it to heat up the high temperature side of Stirling engine combined with generator and get electricity.
- ▶ The other three CSP technologies have heat storage part. They use collector to absorb heat and transfer to the heat conserving fluid, usually oil, or molten salt. Then the fluid is heated to high temperature pumped to the separate tank for storage. When needed, it will be pumped through a fluid-to-steam heat exchanger and produces superheated steam for steam engine to generate electricity.
- ▶ In hybrid systems they would comprise fossil fuel power plant + CSP heat water supporting system.

Principles of Photovoltaic (PV)

- ▶ Traditional PV uses semiconductor materials such as Silicon to fabricate PV cells which generate direct current under sunlight.
- ▶ First PV cell: 1954.
- ▶ Early applications: space for power supply in satellites.
- ▶ It is now commonly used on rooftop for supplementing electricity from utility grid.
- ▶ Details about PV systems shall be given in the coming lectures.



PV physics



https://youtu.be/L_q6LRgKpTw?t=81

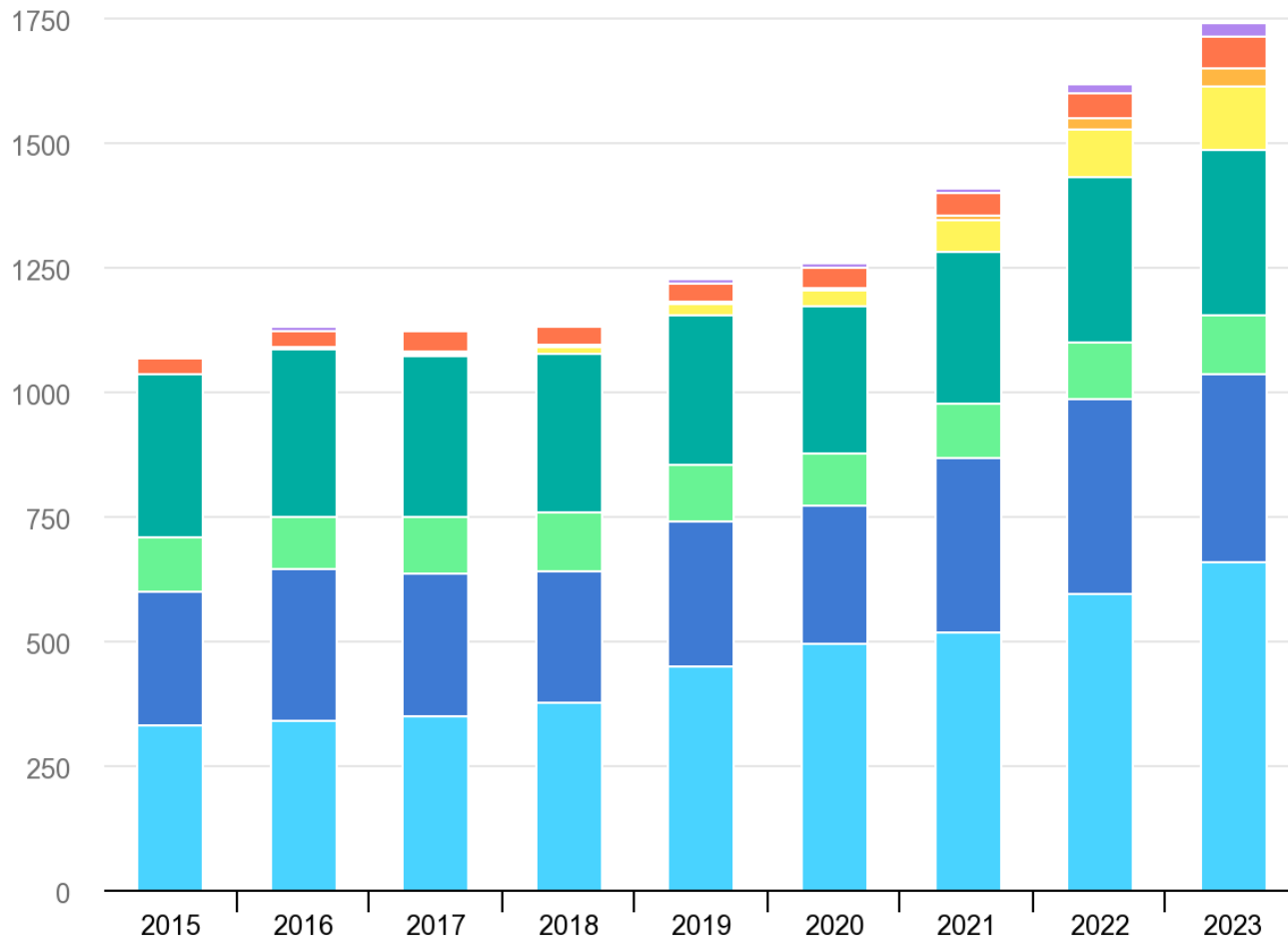
Considerations of applying RE

- ▶ There is no difficulty in developing RE in terms of technology and resources. However social and economical factors should also be considered:
 1. Land use - PV and wind require large area of land which can be expensive in urban area.
 2. Locations of application - Hydropower, Tidal power, and Geothermal are dependent on local resources.
 3. High density of buildings - this would limit solar access which poses limitation in PV application, therefore appropriate city planning, such as protecting the right to have solar access by legislative means is required in long-term.
 4. Visual and noise nuisance - RE installations may be visual and noise nuisance to some residents. E.g. some believe wind turbines produce noise and PV modules produce glare.
 5. Costs of RE are dropping, in particular the large-scale applications. Distributed generation are still comparatively expensive.

Cost and future of RE

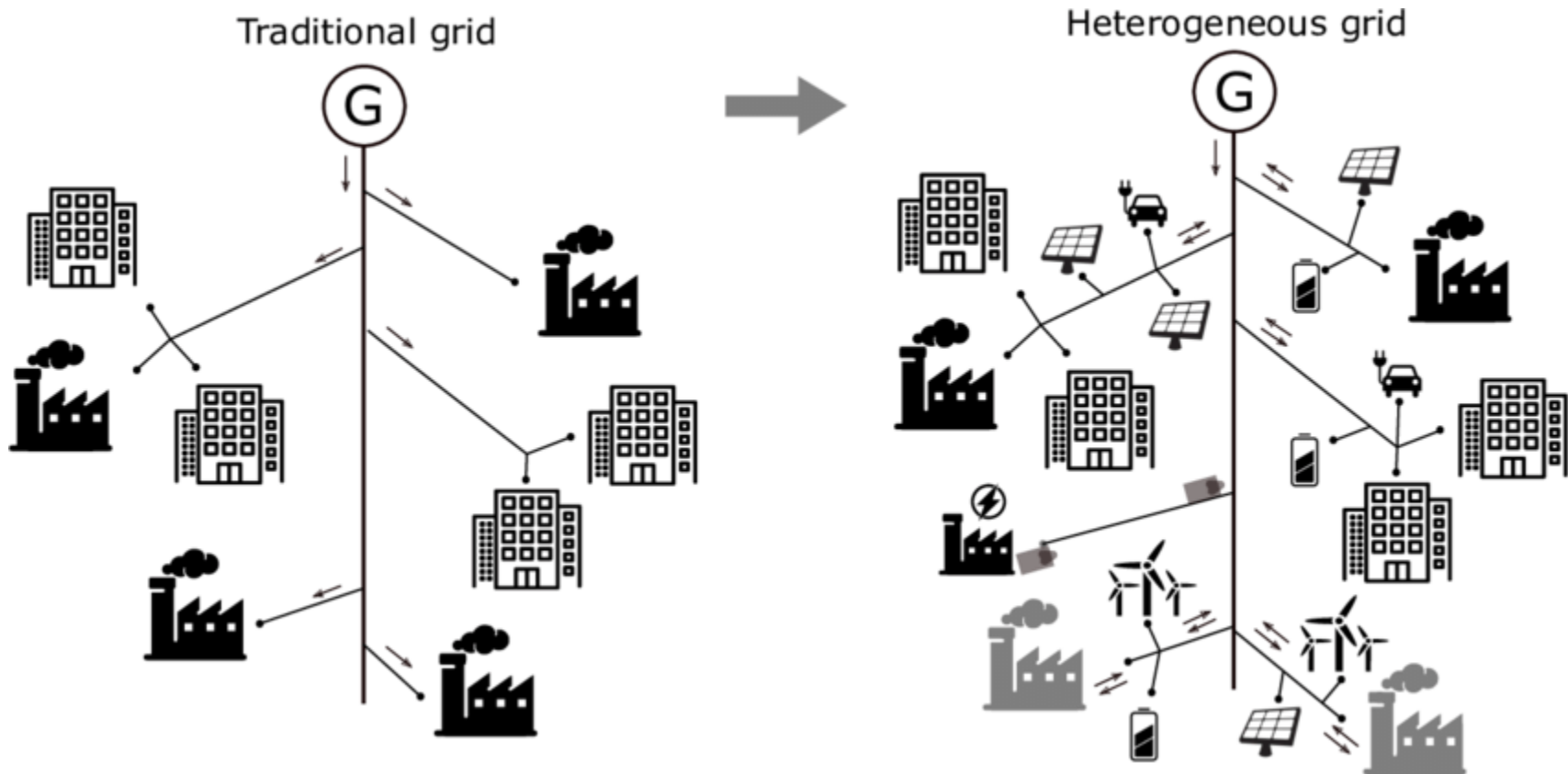
- ▶ Wind, hydro, geothermal, and solar (both thermal and PV) are all cost competitive.
- ▶ More than cost: only RE is possible with definite “*energy pay-back period*”; whereas non-RE cannot achieve any energy pay-back.
- ▶ Enormous resources have been invested to RE research in developed countries such as EU and Japan.
- ▶ Cost per kWh is decreasing as the production capacity of RE equipment increases.
- ▶ Breaking news: “2024 – a Year of Records for Renewables: Unprecedented 14% increase in renewables capacity”.

Global investment trend



Source: <https://www.iea.org/data-and-statistics/charts/annual-clean-energy-investment-2015-2023>

From centralised to decentralised



Source of image: Filipe PEREZ, *“Control of AC/DC Microgrids with Renewables in the Context of Smart Grids: Including Ancillary Services and Electric Mobility”*. Université Paris-Saclay; Universidade federal de Itajuba. Brésil, 2020.

Development of RE

- ▶ Traditional power system is engineered to safely serve a given level of load at a given level of reliability with minimal cost.
- ▶ Electrical power is generated in central power plants and then transmitted (transmission) and distributed (distribution) to various load centres.
- ▶ The radial distribution system is configured similar to a tree, with the feeder outlet being the trunk and the leaves/fruits being the loads, supported by branches of progressively smaller sizes. So, no two circuit are exactly alike.
- ▶ The complexity of modern grid system grow with the number of RE connected and to be in operation with the existing utility grid.
- ▶ Grid infrastructure improvements are required – deployment of smart grid to cater the changes.

The role of smart grid

- ▶ The power output from most of the RE Systems is directly proportional to the natural resources available.
- ▶ E.g. wind speed, solar irradiance, etc. are changing minute by minute, and could differ a lot in short period of time -> the **intermittency**.
- ▶ When the percentage of contribution from the RE systems to the grid is up to a certain level, this would bring a stability issue to the grid.
- ▶ Hence many features, including the capability to forecast RE outputs and load demands by a smart grid would be very useful in maintaining grid stability.
- ▶ Also, when **load shifting** is required to smooth out the electricity demand, smart grid plays an important role when these RE systems with intermittent nature exists.
- ▶ Other smart grid features include: demand response programmes, smart appliances, etc.

Recommended reading

1. IPCC website: <https://www.eipcc.ch/>
2. HKSAR Government EMSD's leaflet about renewable energy:
“Know more about Renewable Energy”
https://www.emsd.gov.hk/filemanager/en/content_299/re_leaflet_eng.pdf
3. R. Wengenmayr and T. Bührke (2013), *Renewable energy: sustainable concepts for the energy change*, Weinheim
<https://onlinelibrary-wiley-com.eproxy.lib.hku.hk/doi/book/10.1002/9783527671342>